

Pandemic Through Time: Probabilistic Modelling of Coping Profile Dynamics

A Research Project by

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This thesis, entitled “**Pandemic Through Time: Probabilistic Modelling of Coping Profile Dynamics**”, prepared and submitted by **Serge Alec M. Rivera**, in partial fulfillment of the requirements for the degree of **Bachelor of Science in Computer Science** is hereby accepted.

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Abstract

Coping is a strategy to manage stress. Given the importance of coping as a predictor for mental health, coping profiles during the COVID-19 pandemic, characterized by mental health, lifestyle conditions, and coping behaviors, were discovered using a k-means clustering model during the first phase of this project. The next phase of the project implemented a temporal analysis of the identified coping profiles to determine how coping profiles and related factors changed throughout the pandemic. The k-means clustering model was reapplied on data (N1=20610; N2=11453; N3=6080; N4=16641) from four mild lockdown periods (Feb 24-28, 2021; Jun 14-20, 2021; Aug 10-23, 2021; May 13-30, 2022) in Japan to label the coping profiles across all surveys. The frequencies of profile transitions were converted to a Markov chain, which models the probabilities of profile transitions. A hidden Markov model was trained on the survey data to compare another Markov chain fitted to the observed data. Statistical tests were conducted to identify factors associated with the profile transitions. The resulting Markov chains had highly similar transition probabilities and stationary distributions, with the predicted distribution of the population being within the Ambivalent, Resilient, and Optimistic profiles. This result was related to the majority of the transitions being accompanied by changes indicating the public adapting to the pandemic: a decrease in perceived shortage of COVID-related supplies and anxiety brought by news about the pandemic, while an increase in vaccination willingness and offline interaction. Struggling financially was identified to be related to Highly Stressed profile transitions. Transitions to the Optimistic profile were related to an increase in income and the coping behaviors: following a healthy diet,

pursuing hobbies, exercising, interacting socially, and thinking positively about the future. The mentioned coping behaviors are recommended for potential improvement of one's mental health state. Government institutions may also want to look into the dynamics of coping profiles for indicators that the general population has adapted to a public crisis.

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Chapter 1

Introduction

1.1 Background of the Study

The global COVID-19 pandemic had a significant impact on the public's mental health brought about by the safety protocols imposed to contain the virus. A study on the mental health of residents in 18 countries within the Middle East and North African (MENA) regions reported that over 40% of the residents were experiencing moderate to severe psychological distress because of the pandemic [1]. In Europe, 8 countries were discovered to have higher stress levels compared to the pre-pandemic levels with over 35% of the sampled population exhibiting moderate to severe depressive symptoms [2]. In Japan, over 49% of samples from seven prefectures were exhibiting social isolation throughout two mild lockdown periods [3]. From these reports, it is essential to find the risk and protective factors of psychological distress during the COVID-19 pandemic in order to plan mental health interventions.

Individuals reacted differently to the pandemic through coping strategies. Adults from Spain coped by following a healthy/balanced diet, not reading news/updates about COVID-19 very often and taking the opportunity to pursue hobbies; these

enumerated coping strategies were associated with lower levels of anxiety and depression [4]. Respondents from the USA and Canada were coping mainly through acceptance and distraction but it was identified that distraction was related to higher depression, higher anxiety and lower quality of life [5]. A study on the general public revealed that avoiding thinking about the pandemic was a coping strategy associated with higher levels of anxiety and depression [6]. There are effective and detrimental coping strategies during the pandemic and numerous research expand further on these strategies and its direction through longitudinal studies and coping transition analysis.

Recent longitudinal research involves identifying coping strategy patterns within a population, commonly referred to as profiles or classes, across different time periods. Transitions to different profiles or classes are computed to understand the significant changes. Additional analyses are done on other variables or factors that could be related to the transitions. Key findings in some longitudinal studies include: more transitions from a negative class to a positive class caused by the pandemic, avoidant coping strategies associated with higher levels of psychological distress, and emotion-focused coping, technology overload, and technology invasion associated with transitions to a negative class [7][8][9]. However, the limitation of these studies is that data measured only involved two time points: before and during the first wave of the COVID-19 pandemic. With the multiple resurgence of COVID-19, research can provide insights on how coping mechanisms developed over the course of the pandemic and possible factors that could have affected said development.

1.2 Objective

The objective of this study is to further explore the coping profiles and understand its trend throughout the pandemic. The study will analyze collected data from different lockdown periods in Japan and provide a graphical model of the coping profile dynamics and factors that could have affected the profile transitions. Findings from this study can be used by government institutions to plan interventions in the case of a long term pandemic similar to the COVID-19 pandemic.

Chapter 2

Review of Related Literature

2.1 Coping Behaviors and Mental Health during the Pandemic

Impact of COVID-19 on mental health and quality of life: is there any effect? a cross-sectional study of the MENA region (Al Dhaheri et al., 2021) [1]

A cross-sectional study was implemented to identify the mental health impact of the COVID-19 pandemic to the population residing in the Middle East and North Africa (MENA) region. Statistical analyses were applied to report any considerable differences related to mental health levels and socio-demographic characteristics.

Respondents from 18 MENA countries (Algeria, Bahrain, Egypt, Iraq, Jordan, Kuwait, Lebanon, Libya, Morocco, Oman, Palestine, Qatar, Republic of Yemen, Saudi Arabia, Sudan, Syria, Tunisia, and United Arab Emirates) during the COVID-19 outbreak, May 2020 to June 2020, were recruited. A total of 6142 responses were received with the researchers aiming for 100 responses from each country.

The survey collected sociodemographic characteristics: age, gender, country of residence, education level, employment status, marital status, and work or study

setting. Furthermore, multiple scales were utilized to assess certain variables: psychological impact with the Impact of Event Scale-Revised Scoring (IES-R), negative mental health impacts with modified questions from a Severe Acute Respiratory Syndrome (SARS) study, social and family support impacts with a modified version of the Perceived Support Scale (PSS), and lifestyle changes with modified questions used in a SARS study.

With respect to the statistical analyses, the Kolmogorov-Smirnov test of normality was first applied to check for the normality of the data. Descriptive statistics were used for the sociodemographic characteristics. Chi-squared test was used on the categorical variables and IES-R responses. Non-parametric Kruskal-Wallis H test with pairwise post-hoc analysis using Bonferroni adjustment was used on the IES-R, intrusion, avoidance, and hyperarousal scores to check for any significant differences in the scores between the MENA regions. Multivariate linear regression was used to check for the relationships across all the measures.

Results showed that the North African regions obtained significantly higher IES-R scores compared to the other regions. Across all MENA regions, 40% of the respondents were suffering from moderate to severe mental distress. Risk factors associated with high levels of mental distress include: females, aged 26 - 35, low education background. With regards to personal matters, more than one third and more than one half of the respondents were being stressed because of work and financial matters and home matters, respectively. 40% of the respondents were focusing on their mental health and spending the time to rest. For coping mechanisms, majority of the respondents coped through support from family members; however, 42% of participants spent less time on physical exercise.

In summary, the study presented the severity of mental health across all MENA regions. Proponents of the study challenged government institutions to implement interventions that would target the significant variables presented: risk factors, stressors, and coping mechanisms.

Mental health among medical professionals during the COVID-19 pandemic in eight European countries: cross-sectional survey study (Hummel et al., 2021) [2]

The study made a comparison between medical and nonmedical professionals in eight different European countries during the COVID-19 pandemic with the objective of distinguishing the factors affecting mental health and coping strategies of the subjects. Statistical analyses were implemented to detect any notable changes.

Subjects were 354 medical and 255 nonmedical professionals residing in Germany, the United Kingdom, Spain, France, Portugal, Austria, Italy, and Switzerland. Surveys were distributed during peaks of the pandemic which ranged from April to June 2020.

Data collected include demographic profile, mental stress measured using the Depression Anxiety Stress Scales-21 (DASS-21), stressors which were only asked to medical professionals, and coping strategies.

Comparison of means, standard deviation and frequencies and multivariate analysis of variance were the statistical tests applied to conclude significant differences between the two demographics.

Results showed that one third of medical professionals were suffering from moderate to severe depression, anxiety and distress. However, depression and anxiety levels were significantly lower among the medical professionals than the nonmedical professionals. No significant difference between direct contact with COVID-19 patients and

the DASS scores was identified among the medical professionals. “Uncertainty about when the epidemic will be under control” and “worry about inflicting COVID-19 on family” were the top stressors of the medical professionals while “taking protective measures (washing hands, wearing a mask, taking own temperature),” “actively acquiring more knowledge about COVID-19 (symptoms, transmission pathway, etc),” and “video-chatting with family and friends to share concerns and support” were the top coping strategies of the medical professionals. Psychological distress levels were discovered to be higher than the levels prior to the pandemic. Respondents from the United Kingdom and France had the highest levels of which may be attributed to the high transmission and fatality rates in said countries.

In short, the study was able to present that a substantial ratio of medical professionals from different European countries were suffering from mental health distress during the COVID-19 pandemic. Prominent stressors and effective coping strategies were also enumerated in the study.

The transition of social isolation and related psychological factors in 2 mild lockdown periods during the COVID-19 pandemic in Japan: longitudinal survey study (Sugaya et al., 2022) [3]

The study conducted two online surveys during COVID-19 outbreaks in Japan to examine social isolation transitions. Through clustering and statistical tests, relationships between the transition patterns of social isolation and psychosocial factors were analyzed.

Respondents were composed of residents from locations where the first emergency measures were implemented. The first survey collected 11333 responses in May 2020; while the second survey, a follow-up to the first survey, collected 7893 responses in February 2021.

Questions in both surveys include:

1. sociodemographic data which is composed of age, sex, employment status, marital status, annual household income, number of cohabitants, whether the individual or a family member was a health care worker, being treated for a mental condition or severe physical disease, and had a history of treatment for a mental disorder or severe physical illness
2. social isolation measured with Lubben Social Network Scale (LSNS-6)
3. loneliness measured with UCLA Loneliness Scale, Version 3 (UCLA-LS3)
4. psychological distress measured with Kessler Psychological Distress Scale-6 (K-6)
5. Lifestyle, coping behavior, and stressors related to mild lockdown obtained using questions developed by the proponents of the study

Respondents were classified into social isolation groups based on their LSNS-6 scores:

1. No SI group - no social isolation on both surveys
2. Improved SI group - had social isolation in the 1st survey but not in the 2nd survey
3. Worsened SI group - no social isolation in the 1st survey but had in the 2nd survey
4. Persistent SI group - had social isolation on both surveys

Statistical comparisons were made between the sociodemographic characteristics and the psychological measures through the chi-squared test and t-test. Sociodemographic data between the four social isolation groups were compared using a chi-squared test. Mild lockdown stressors and psychological measures between the social isolation groups and responses from different surveys were analyzed using two-way ANOVA. To visualize the relationship between the transition patterns of social isolation and psychosocial factors nonparametric Bayesian co-clustering was performed.

Statistical results showed no significant improvements in the LSNS-6 and UCLA-SC3 scores. On the other hand, there were significant improvements in the K-6 scores and slight reduction in depression. With respect to social isolation, 49% of the respondents are in the Persistent SI group while 12% are in the Worsened SI group. The analysis discovered that respondents belonging in the Persistent SI group were either unmarried or had no children but generally had fewer cohabitants than the other social isolation groups. Additionally, low-income respondents were more prevalent in the Persistent SI group.

The clustering output confirmed that most clusters with No SI group members exhibited healthy behaviors, more interactions, good relationships, less loneliness, psychological stress and were able to fix their relationships; while clusters with Persistent SI group members may belong to two subclusters: increased loneliness and psychological stress subcluster and close to respondents' average scores subcluster. The former subcluster was subject to deteriorating relationships and less interactions.

Overall, the study reported that social isolation was prevalent in Japan between two time periods during the COVID-19 pandemic. Key psychosocial factors and demographic data were linked to social isolation transitions and can be utilized for

intervention planning.

Coping behaviors associated with decreased anxiety and depressive symptoms during the COVID-19 pandemic and lockdown (Fullana et al., 2020) [4]

The study conducted a survey with the goal of showing the effectiveness of coping behaviors to anxiety and depression during the COVID-19 pandemic

Participants of the study included 4399 adults from the Spanish population. The survey was done two weeks after a lockdown period.

The survey collected sociodemographic characteristics, labor status, anxiety levels measured with the Generalized Anxiety Disorder-7 (GAD-7), depression measured with the Patient Health Questionnaire-9 (PHQ-9), risk of COVID-19 for self and close relatives, exposure to life-events not associated with COVID-19 and frequency of coping behaviors the past two weeks.

Ordinal regressions were performed to identify the relationship between the coping behaviors, GAD-7 and PHQ-9 scores. The scores were set as the dependent variables while the frequency of the coping behaviors as the independent variables.

It was discovered that 65% of the respondents had anxiety or depressive symptoms. Moreover, findings show that lower levels of anxiety and depression were related to the coping behaviors: following a healthy/balanced diet, not reading news/updates about COVID-19 very often, and taking the opportunity to pursue hobbies. Lower anxiety was related to the coping behavior: not talking to relatives/friends very often. On the other hand, lower depression was associated with the coping behaviors: following a routine, and staying outdoors or looking outside.

Given the discovered relationship between anxiety, depression and coping behaviors, health institutions can use the identified coping behaviors to alleviate mental

distress that came with the COVID-19 pandemic.

Coping during the COVID-19 pandemic: relations with mental health and quality of life (Shamblaw et al., 2021) [5]

In the study, different coping behaviors were analyzed along with quality of life and symptoms of depression and anxiety during the COVID-19 pandemic. The objective of the study is to determine the cross-sectional and longitudinal relationships between the three factors.

The study gathered 797 respondents from the USA and Canada who answered a survey on April 21 to 25, 2020. A follow-up survey was sent on May 21 to 27, 2020 but only 395 responses were collected for the follow-up.

The surveys provided collected the coping strategies of the respondent through Brief Coping Orientation to Problems Experienced (Brief COPE), quality of life through World Health Organization Quality of Life-BREF (WHO-QOL-BREF), depressive symptoms through the Patient Health Questionnaire-9 (PHQ-9), anxiety symptoms through the Generalized Anxiety Disorder-7 (GAD-7), COVID-related context, and questions to filter out respondents.

To determine the relationship between coping strategies, quality of life, and depressive and anxiety symptoms, linear regression analysis was performed. As for the mediating effect of the symptoms between coping strategies and quality of life, mediation analysis was implemented.

Results of the analyses showed that positive reframing and active coping were related to lower levels of depression and higher quality of life. Positive reframing was also beneficial to quality of life even after considering the follow-up survey. Coping through planning was related to higher levels of anxiety. Self blame was identified to

be the only maladaptive coping strategy which is related to changes in mental health and quality of life. As for the mediation analysis, the relationship between avoidance coping set of strategies and quality of life was mediated by anxiety and depression. The most common coping strategies were acceptance and distraction but distraction was associated with higher depression, higher anxiety and lower quality of life.

The study showed that coping strategies during the COVID-19 pandemic have significant relationships with depressive and anxiety symptoms. Coping through positive reframing was recommended in future interventions while coping through distraction should be discouraged.

2.2 Coping Profile Transitions

Changes to the work–family interface during the COVID-19 pandemic: examining predictors and implications using latent transition analysis (Vaziri et al., 2020) [7]

The research studied transitions of work-family conflict and enrichment profiles among employees before and during the COVID-19 pandemic. The research goal is to identify factors which could have affected the transitions and understand its implications. There are two parts in the study: (1) identifying the profiles and (2) applying and analyzing the profile transitions between two time points.

The first part of the study involved 379 responses from employees living in the United States (US) and worked at least 32 hours per week. These responses were collected before the pandemic. The second part of the study analyzed 583 and 371 responses from employees living in the US and worked at least 32 hours per week during the second week of February 2020 and first week of April 2020, respectively. These dates correspond to before and during the COVID-19 crisis in the US.

For the measures in the surveys, the first part of the study measured work-to-family conflict, family-to-work conflict, work-to-family enrichment, and family-to-work enrichment. In addition to the measures of the first part of the study, the second part included the following measures: personal characteristics, segmentation preference, problem-focused coping, emotion-focused coping, technology overload, technology invasion, technology incompatibility, family-supportive supervisory behaviors, leader compassion, job satisfaction, organization commitment, turnover intentions, organization citizenship behaviors and job performance. Responses to these questions range from 1 (strongly disagree/never) to 5 (strongly agree/very often).

The first part of the study utilized latent profile analysis to identify the optimal number of profiles to cluster the responses from the first survey. The second part combined latent profile analysis and latent transition analysis to apply the optimal number of profiles in the two COVID-19 surveys then determine the probabilities of profile transition between the two surveys. Paired sample t-test was implemented to check for significant factors between the respondents who experienced profile transition.

Results from the first part of the study suggest three profiles which were also applied in the second part of the study. These three profiles are beneficial (low conflict and high enrichment), active (medium conflict and enrichment), and passive (low conflict and enrichment). From the transition probabilities, it was concluded that passive profiles were more likely to transition to the active profile implying the worsening or improvement of conditions during the height of the pandemic. Moreover, those with beneficial profiles tend to stay within this profile. Transitions from beneficial to active profile were related to those with higher levels of emotion-focused

coping, technology overload, and technology invasion. High levels of leader compassion and lower segmentation preferences were associated with transitions from active to beneficial profile. Active to passive profile transition was observed among those with lower levels of family-supportive supervisory behaviors.

In conclusion, the study was able to identify profiles associated with work-family conflict and enrichment during the COVID-19 pandemic. Likelihoods of profile transitions and factors associated with profile transitions were presented and can be used by employers to alleviate some of the stresses caused by work-life changes resulting from a pandemic.

Hidden in plain sight? men’s coping patterns and psychological distress before and during the COVID-19 pandemic (Livingston et al., 2020) [8]

A study on men’s coping patterns before and during the COVID-19 pandemic was conducted to achieve three goals: identify coping profiles, relationships between the coping profiles and psychological distress (stress, anxiety, depression, and anger) symptoms, and relationships between coping profiles and appraisals of pandemic-related stressors and coping options.

The study was composed of two data collections before and during the COVID-19 pandemic. The first data collection was done from July 2017 to July 2019 and surveyed 409 men from Australia; while the second data collection was done from March 21 to May 19, 2020 with over 272 respondents.

Measures of the study included the following:

1. Stress, anxiety and depression were measured using the Depression Anxiety Stress Scale (DASS-21)

2. State anger was measured using the state-trait anger expression inventory (STAXI-2)
3. Use of coping strategies was measured using the 28-item Brief Cope
4. Appraisals of pandemic-related stressors and coping options were measured using a survey based on the Cognitive Appraisal Health Scale
5. Sociodemographic characteristics which are potential confounders were also included in the surveys.

Analyses in the study include multiple latent profile analyses to determine the coping profiles among the respondents. Multiple linear regression were implemented to determine a significant relationship between the coping profiles and symptoms of psychological distress which was used to predict psychological distress symptom levels. Potential confounders, cognitive appraisals, and previous psychological distress symptoms were also used for the prediction.

Three profiles were derived using the latent profile analyses on both time points: Relaxed Copers (low and balanced use of coping strategies), Dual Copers (high use of avoidant and moderate to high use of approach-oriented coping strategies) and Approach Copers (approach-oriented coping strategies). Higher levels of psychological distress symptoms were observed among those who avoided or distracted themselves from COVID-19 stressors including Dual Copers. Relaxed and Approach Copers had lower levels of psychological distress symptoms with acceptance, self-distraction, planning and active coping as their coping behaviors. It was discovered that most men changed their coping behaviors during the pandemic to be more relaxed and more approach-oriented which was associated with lower levels of psychological distress and

stronger resilience against the pandemic. On the other hand, those who retained or introduced avoidant coping strategies including Dual Copers were subject to higher levels of psychological distress, threatened and overwhelmed by the pandemic, and saw less coping options.

In summary, the analyses before and during the COVID-19 pandemic described that men who followed approach-oriented and less coping strategies were more likely to have a healthier mental state than men who adopted or developed avoidant coping. Of the coping profiles identified, Dual Copers were at risk of psychological distress despite them also practicing approach-oriented coping strategies.

Resilience patterns of Swiss adolescents before and during the COVID-19 pandemic: a latent transition analysis (Janousch et al., 2022) [9]

A study on adolescents from Switzerland was conducted to find any changes in resilience classes before and during the COVID-19 pandemic. As observed in previous studies, the researchers hypothesized that there will be three to four resilience classes, changes in resilience classes after the start of the COVID-19 pandemic and relationship between gender and migration background and the identified resilience classes.

The study had 317 Swiss students to participate in the longitudinal study. The first data collection was performed in August to September 2019 and the second data collection in August to September 2020 when the COVID-19 pandemic spread further.

Age, gender and migration background were requested in the surveys. Psychological distress which is composed of anxiety and depression were measured using the Hopkins Symptom Checklist-25 (HSCL-25). Resilience, which is composed of personal competence, social competence, structured style, social resources, and family cohesion subscales, was measured with the Resilience Scale for Adolescents.

T-tests were used with descriptive statistics to check for any significant differences between the two time points. Latent class analyses were used separately on the two time points to generate the resilience classes. Latent transition analysis was performed to study the changes in resilience classes between the two time points. R3STEP was implemented to check if the gender and migration background can be used to predict membership in a resilience class.

Three resilience classes were derived by the latent class analysis: Resilient, Non-resilient and Untroubled. The Resilient class was characterized by high protective factors (personal competence, social competence, structured style, social resources, and family cohesion) and high mental health issues (anxiety and depression). Non-resilient class had low protective factors and high mental health issues. Untroubled class had high protective factors and low mental health issues.

Findings show that depression levels were higher during the pandemic compared to before the pandemic but anxiety levels remained the same. Social competence decreased in the Nonresilient and Untroubled classes but increased in the Resilient class. However, social resources and family cohesion increased for all classes but not as much in the Nonresilient class. Personal competence slightly increased in the Resilient and Nonresilient classes. Structured style increases in all classes.

In the latent transition analysis, most of the classes were stable, i.e. less class transitions were observed, with 70% stability in Nonresilient and Untroubled classes and 90% stability in the resilient class. Many students in the Untroubled class transitioned to the resilient class implying that students still had high protective factors but the pandemic may have caused higher mental health issues. During the pandemic, most of the students were in the Resilient class and girls were also discovered to be

more likely to be in the Resilient and Nonresilient classes which has the implication that girls were more prone to mental health issues during the pandemic.

To address the hypotheses of the research study, three classes were the optimal number of classes to classify resilience patterns derived through latent class analysis, class stability was more prevalent but some class transitions with significant implications were observed in the study through latent transition analysis, and being female was a risk factor to worse mental health during the COVID-19 pandemic.

2.3 Markov Models in Mental Health Studies

Modeling depression progression dynamics from electronic health record (Huang, 2016) [10]

The research modeled changes in depression severity through b-spline curving, k-means clustering, and Markov models in order to study and predict the progress of patients suffering from depression.

Subjects of the study were patients which were receiving treatment for 180 days. Data gathered was aggregated into two weeks, i.e. values obtained in two weeks were averaged. Subjects consist of 6067 patients with fewer than four responses per biweek and 610 patients with more than six responses per biweek. Data from the two types of patients were separated and compared throughout the study.

The responses collected were the five severity states devised by the Patient Health Questionnaire-9 (PHQ-9): healthy, mild, moderate, moderately severe and severe.

A b-spline curve was fitted to the dataset in order to predict the trajectory of the data given that the dataset had missing values and irregular time of responses. K-means was applied to create the clusters based on the trajectory derived from the

b-spline curving. Four Markov models to simulate the severity transitions were devised in the study: discrete-time Markov model, hidden Markov model, semi Markov model, and hidden semi Markov model. The stationary probabilities, from the Markov models, of each cluster, from the k-means clustering, were analyzed and compared.

Discussing each Markov model, the discrete-time Markov model is the baseline Markov model wherein the transition states of depression severity were calculated for each biweekly transition from baseline to the final biweek of a certain patient. The hidden Markov model relied on the transition states probability from the discrete-time Markov model for the state transition matrix and an existing study's severity distribution for the emission or output matrix. The hidden Markov model was applied because in real world applications only scores are observed but actual diagnosis severity of depression is uncertain. The semi Markov model was applied to consider the amount of time a patient stays in a certain state. The hidden semi Markov model combines the hidden Markov model and semi Markov model assumptions. All of these Markov models were simulated in the long run for the stationary probability distribution.

Results of the clustering process showed that the selected number of clusters for both datasets is five which correspond to trajectories of the PHQ-9 scores namely, mild, increasing severity, decreasing severity, moderate, severe. The increasing severity and severe clusters were discovered to be not improving over time with treatment.

After simulating the Markov models, it was shown that the 6067 patients in the increasing severity cluster had higher probabilities of being in the moderately severe and severe depression states than the 610 patients. Moreover, it would take fewer than 20 biweeks and 35 biweeks for patients in the increasing severity cluster to be

in moderately severe and severe depression states, respectively. In the semi Markov models, it was identified that the mild and decreasing severity clusters spent more time in the healthy state while the severe, moderate, and increasing severity clusters spent more time in the moderately severe and severe states.

In summary, analysis was performed on different clusters, generated using k-means clustering and b-spline curving. The analysis provided the likelihood of depression severity state transitions of patients through Markov models.

Discrete state analysis for interpretation of data from clinical trials (Sugar et al., 2004) [11]

A research was made on schizophrenic patients with the goal of comparing two different medical treatments through k-means clustering, Markov models, and statistical tests in analyzing treatment of refractory schizophrenia.

Respondents of the study were composed of 423 patients from 15 health centers. The patients were administered clozapine and haloperidol which are medications for schizophrenia. The data was collected for six time periods: baseline, 6 weeks, 3 months, 6 months, 9 months and 12 months.

To determine the effects of the treatment, mental health and extrapyramidal medication side effects were recorded. Mental health symptoms were measured using the Positive and Negative Syndrome Scale (PANSS). Extrapyramidal medication side effects were measured using three measures: Abnormal Involuntary Movement Scale (AIMS), Barnes Akathisia Scale (BAS) and Simpson-Angus Scale (SAS). Higher scores in these measures imply more severe symptoms.

Principal component analysis was performed on the two major measures, mental health and side effects, to reduce the number of dimensions of the dataset. K-means

clustering was done on the divided dataset, mental health and side effects scores, in order to cluster the health states of the patients. Cluster mean and profile plots were utilized to determine the necessary number of clusters. Chi-squared test of independence was applied on the cluster states and medication. Transition state probabilities, components of a Markov Chain Model, were calculated for the prediction of cluster state transitions using long-run analysis. Permutation tests developed by the researchers were applied on stationary distributions, probability distribution that will remain constant over time, and groups of receiving clozapine and haloperidol to determine any significant differences over time. Linear regression was used on the duration of treatment within a cluster to predict cost of treatment and the cluster states were mapped with utilities or QALY weights. Following these procedures, permutation tests were applied to check for any significant differences in cost and utilities.

Explained overall mental health, contrast between negative and positive symptoms, subjective emotional distress, hostility, and thought disturbances were characterized by five principal components. As for extrapyramidal effects, four principal components related to overall severity, contrast between akathisia and tardive dyskinesia, extrapyramidal syndromes excluding akathisia, and contrast between facial and extremity movements were generated.

Seven mental health clusters and six side effects clusters were identified by the k-means clustering algorithm. The seven mental health clusters correspond to: (1) mild symptoms (MS), (2) moderate symptoms and high global subjective distress (MS + HGSD), (3) moderate symptoms with high grandiosity (MS + HG), (4) severe negative symptoms with low subjective distress (SNS + LSD), (5) severe negative

symptoms with high subjective distress (SNS + HSD), (6) severe positive symptoms (SPS), and (7) severe symptoms with low subjective distress (SS + LSD). On the other hand, the side effects clusters have the notable characteristics: (1) no side effects (NSE), (2) mild tardive dyskinesia (MTD), (3) mild akathisia (MA), (4) extrapyramidal symptoms (EPS), (5) frank tardive dyskinesia (FTD), and (6) severe tardive dyskinesia and severe akathisia (STD + SA)

The chi-squared tests showed that side effects cluster states were significantly different between the two medications, clozapine and haloperidol, across all time periods except for the baseline while mental health cluster states only experienced significant differences during the 6th and 9th month data collection. In the side effects clusters, there were 20% more clozapine patients in the NSE cluster across all time points then three months after, there were 10% less clozapine patients in the MA cluster. With respect to mental health clusters, clozapine was proven to be more effective at reducing SPS and SS + LSD but may have been a cause of transitions to SNS + LSD later into the trials.

State transition probabilities for three months were calculated for the Markov Chain Model. In the long-run analysis, it was discovered that in the side effects clusters 61% of clozapine patients will be in the NSE cluster while only 38% of haloperidol patients will be in the NSE cluster. For the mental health clusters, there were 4% more patients in the MS cluster, 6% more in SNS + LSD, and 12% less in SNS + HSD and SS + LSD clusters. The ratio of patients in the 12-month and stationary distribution of the long-run analysis were found to be approximately identical implying that the 12-month survey was identical to the long-run.

For the costs, it was noted that clozapine was more expensive but in the long

run will be more cheaper since having less symptoms also suggest less costs. With regards to utility score, clozapine patients scored marginally higher than haloperidol patients.

All in all the research was able to show that clozapine is the better treatment in terms of efficacy, costs and utility through k-means clustering, Markov Chain long-run analysis and statistical tests.

2.4 Theoretical Framework

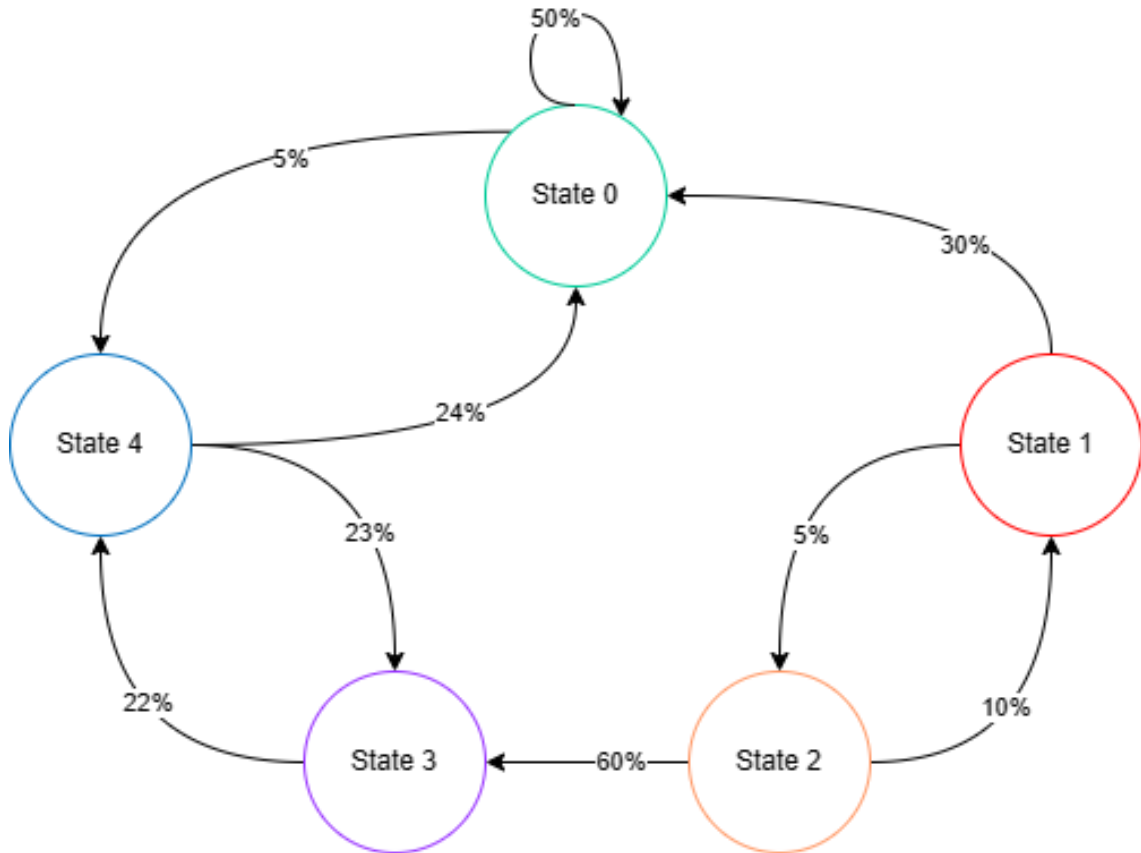


Figure 2.1: Markov chain with five states

A Markov chain is a graphical model with edges representing the probability of the event that a state, represented by a vertex, will transition to another state [12][13][14].

A visual representation of this model is provided in Figure 2.1. This model assumes that the state transitions are independent of the previous states, in other words: the future y_{t+1} is independent of the past $y_1, y_2, y_3, \dots, y_{t-1}$ given the present y_t . This assumption can be relaxed to allow for past states affect the probability of the next state. The state transition probabilities are stored in a matrix which can be simulated through the Markov chain Monte Carlo simulation or a random walk of the model based on the state transition probabilities. Performing this random walk for a significantly long time will yield the stationary distribution of the states.

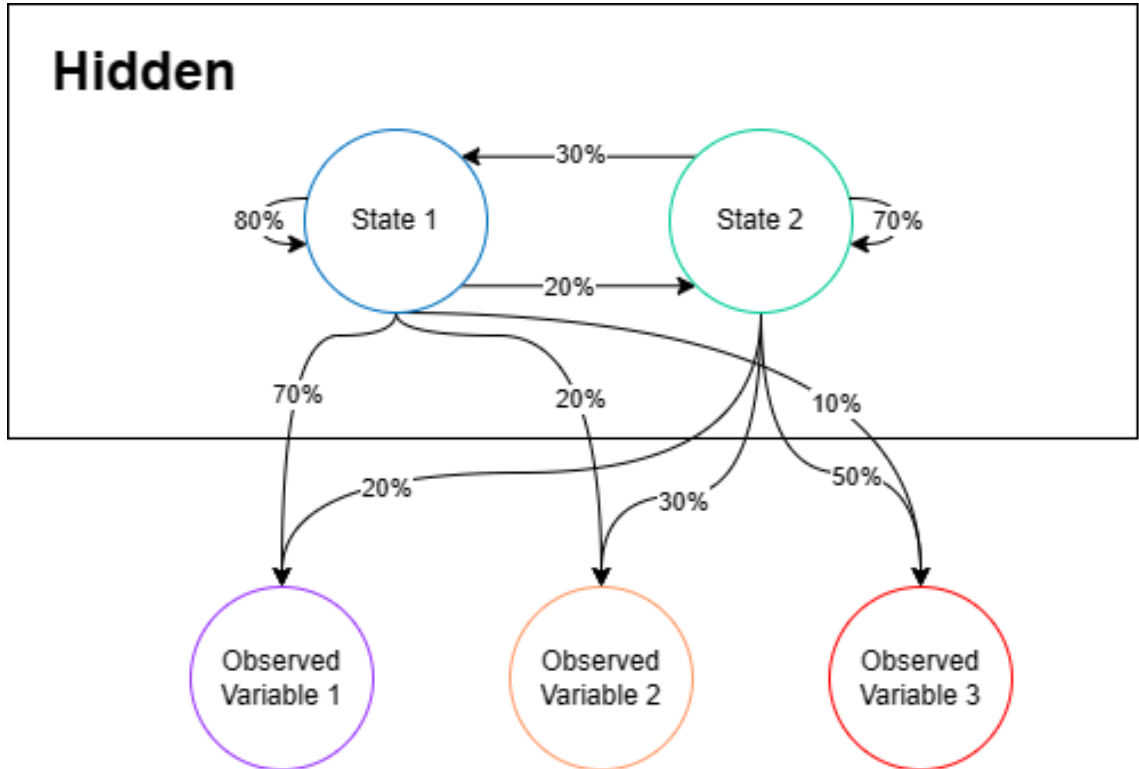


Figure 2.2: Hidden Markov model with two hidden states and three observed variables

The hidden Markov model (HMM) is an extension of the Markov chain. In this model, the Markov chain is hidden and instead the presence of observed variables

determine the state transition probabilities of the hidden states [13][15]. An example of model is presented in Figure 2.2. The model has a state transition probability matrix for the hidden Markov chain and emission probabilities for the probability of an observed variable given a certain hidden state. There are three problems that are solved by the HMM [16][17][13]:

1. Likelihood - Given the observed data, state transition and emission probabilities, determine the probability of the observed data.
2. Decoding - Given the observed data, state transition and emission probabilities, determine the most probable sequence of states.
3. Learning - Given the observed data, determine the state transition and emission probabilities that maximize the probability of the observed data.

For this project, the third problem is most applicable provided that the project's objective is learning how the coping profiles transition to a different coping profile over time. The coping profile dynamics would be discovered using the state transition probabilities learned by the HMM given the observed temporal data. The hidden states correspond to the coping profiles which were discovered by the k-means clustering algorithm in the previous phase of this project while the observed variables are the survey responses.

Chapter 3

Methodology

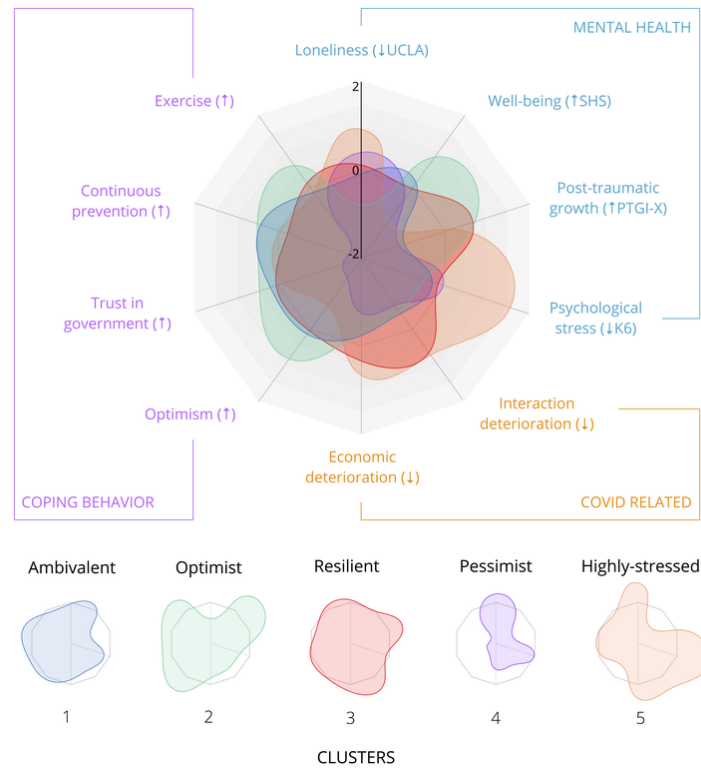


Figure 3.1: COVID-19 coping profiles and their 10 defining features [18]

In the previous phase of this project, sociodemographic, COVID-related lifestyle changes, mental health and coping behavior data were collected from the Japanese

population during the COVID-19 pandemic. A k-means clustering model was applied on the collected data to identify the five coping profiles: Ambivalent, Optimistic, Resilient, Pessimistic, and Highly Stressed [18]. These coping profiles were defined by the features presented in the radio plot in Figure 3.1.

Individuals who belong in the Ambivalent coping profile were not at risk of mental health issues but they did not experience significant post-traumatic growth. The Optimistic profile has the highest optimism scores, trusts the government the most, and has high continuous prevention scores. Optimistic respondents experienced the most post-traumatic growth, highest well-being, and lowest loneliness scores. The Resilient profile experienced high post-traumatic growth despite the high economic and interaction deterioration. Pessimistic individuals scored the least in all coping behaviors and scored the lowest in post-traumatic growth. The Pessimistic profile did not encounter significant economic and interaction deterioration. Highly Stressed individuals had the highest loneliness, highest psychological stress and lowest well-being scores.

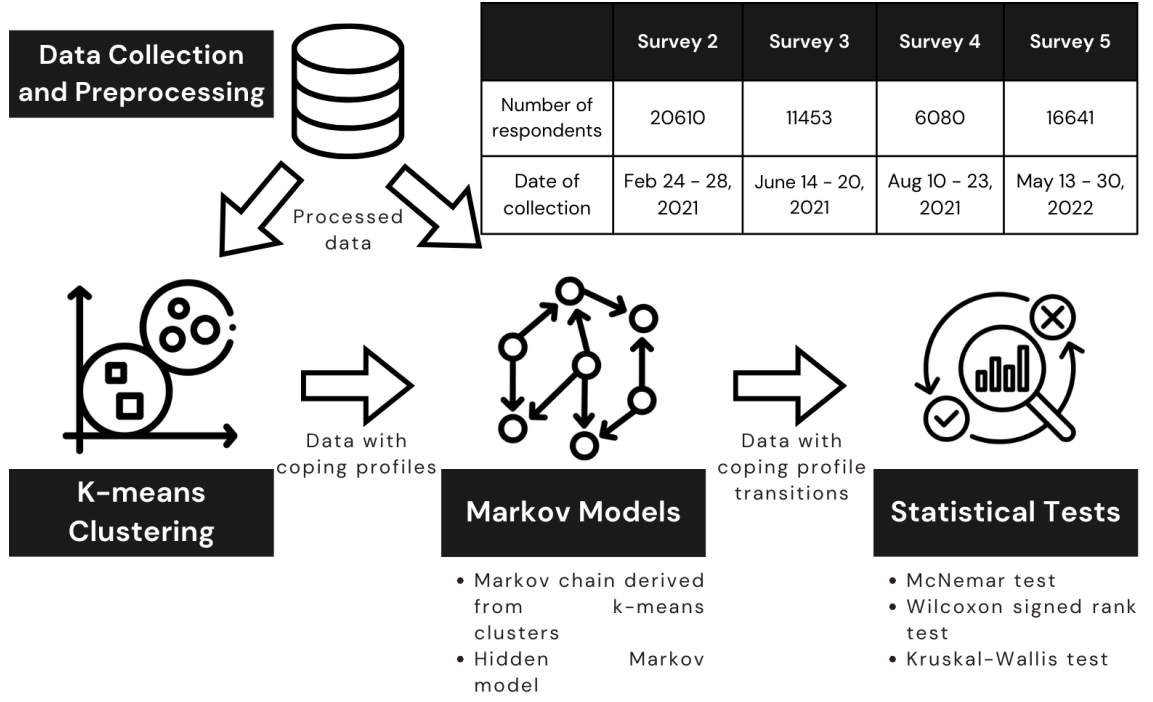


Figure 3.2: Main steps of the methodology

These coping profiles were discovered using data in May 2022. To account for the dynamic nature of humans, a temporal analysis using more time points during the COVID-19 pandemic was implemented in the current phase of the project. The project's methodology is composed of data collection and processing, k-means clustering, Markov models, and statistical tests as indicated in Figure 3.2.

3.1 Data Collection and Preprocessing

The study used data from four surveys (N1=20610; N2=11453; N3=6080; N4=16641) conducted during different mild lockdowns (February 24 - 28, 2021; June 14 - 20, 2021; August 10 - 23, 2021; May 13 - 30, 2022) in Japan. There was a fifth survey collected in May 2020 but it was not included because it was missing too many significant

features.

Features present in these surveys are classified into four categories: sociodemographic, COVID-related, mental health, and coping behavior. Sociodemographic features are made up of general and personal lifestyle characteristics of the respondent. COVID-related features consist of lifestyle changes and reactions to the pandemic. Mental health features are composed of measures that may diagnose a person with mental illness. Coping behavior includes activities which the respondents may have included in their lifestyle to cope with the pandemic. Examples of these features along with their descriptions are presented in Table 3.1.

The preprocessing steps followed for the k-means clustering on the most recent survey was reapplied on the other surveys [18]. Numerical variables were standardized while categorical variables were one-hot encoded. Outliers, composed of respondents who have extremely high or low income, were omitted from the dataset. Given that the surveys did not have the same features, some columns of the other surveys were dropped. Columns which were not present in the other surveys were constructed by imputing data equal to the average values from the most recent survey.

Table 3.1: Feature descriptions

Feature	Description
Sociodemographic	
Prefecture	Prefecture in Japan where the respondent resides
Job/Student	Classification group of job (e.g. government, corporate) or education (e.g. elementary, high school)
Income	Classification group of household and personal income (e.g. less than 2 million yen, between 2 and 4 million yen)
COVID-related	
COVID history	Past medical treatment of COVID
Economic deterioration	Economic situation worsened because of the pandemic
Relationship deterioration	Relationship with close friends and family declined due to the pandemic
Work difficulty	Changes due to the pandemic interfered with work or studies
Mental health	
UCLA loneliness scale	Measure of loneliness
PTGI-X scale	Measure of post traumatic growth
SHS	Measure of well-being
K6	Measure of psychological stress
Coping behavior	
Exercise	Amount of exercise during the pandemic
Healthy diet	Frequency of eating a nutritionally balanced meal
Healthy sleep	Sleep schedule was unaffected after the pandemic
Optimism	Confidence in a positive future despite the pandemic

3.2 K-means Clustering

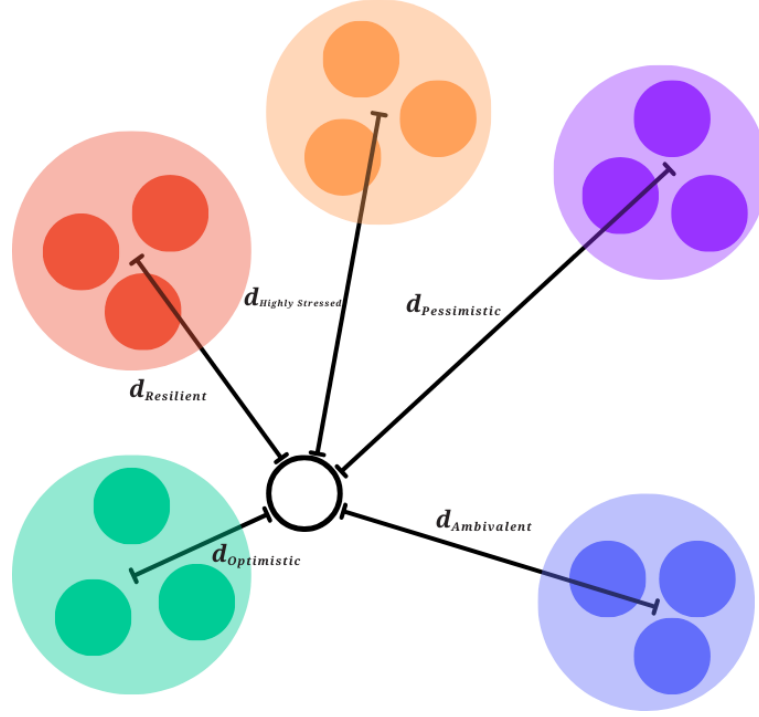


Figure 3.3: K-means clustering model as a classifier

Using the k-means clustering model trained on the most recent survey [18], the k-means clustering model was used as a classifier to predict the coping profiles of the respondents in the other surveys. This classification is done by computing the distances between the learned clusters and the new data [19]. The new sample will be classified in the cluster where its distance between a cluster's centroid is shortest. Mathematically, this is represented as:

$$\text{Predicted Cluster} = \arg \min(d(C)) \quad (3.2.1)$$

$$d(C) = \sum_{i=1}^{\# \text{ of features}} (x_i - c_i)^2 \quad (3.2.2)$$

where

C = cluster; x = new sample; c = cluster's centroid

To validate the classification results of the k-means clustering results, radio plots of the 10 defining features were compared with the radio plot discovered using the most recent survey which was presented in Figure 3.1. A visually resembling radio plot would indicate that the same coping profiles were discovered in the other surveys.

3.3 Markov Models

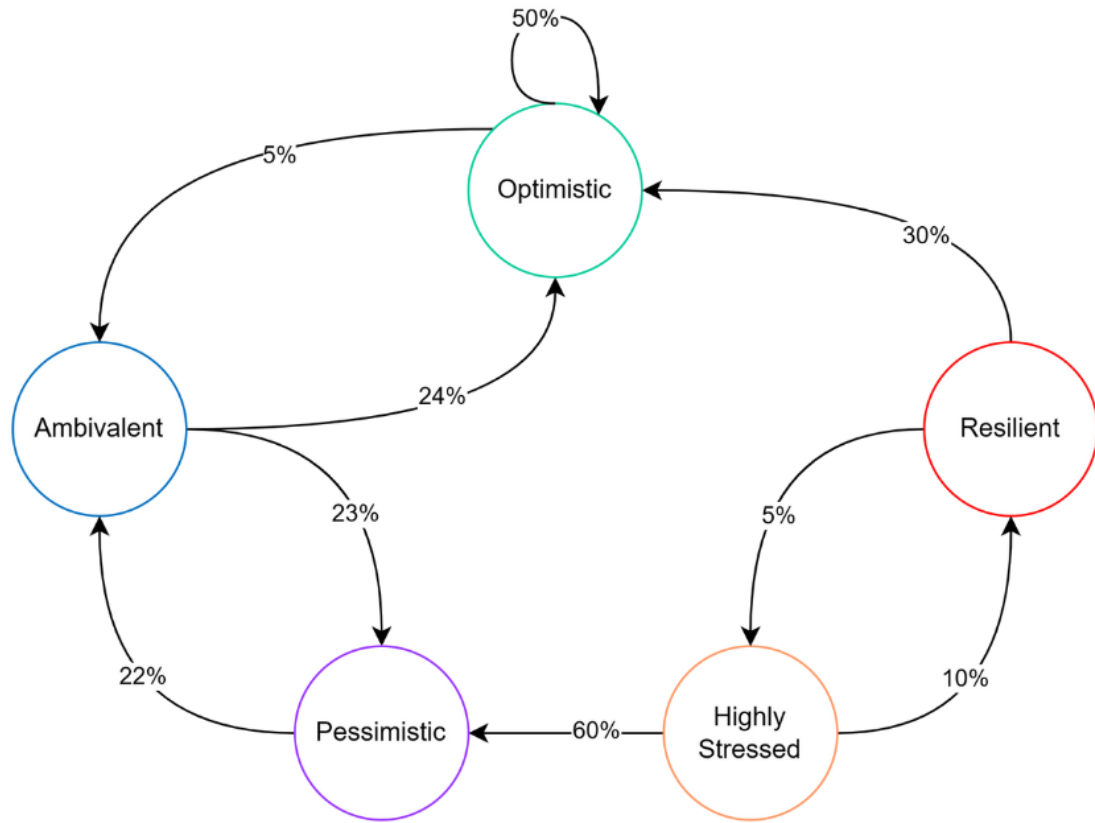


Figure 3.4: Sample Markov chain with coping profiles as states

Markov models were implemented to capture the probabilities of an individual's coping profile developing into a different coping profile given the dynamic nature of humans. These models assume that the (1) coping profiles are the discrete states of the Markov models, represented by an example in Figure 3.4, (2) a respondent must

belong in only one coping profile, and (3) the succeeding coping profile is only dependent of the present coping profile [12][13][14]. Two Markov models were compared: a k-means clustering derived Markov chain and a hidden Markov model's learned Markov chain.

The result from the classification of the k-means clustering model is data with labelled coping profiles. Using the chronological order of the surveys, coping profile transitions, which are paired sequences of data with coping profiles, were recorded. From the frequency of coping profile transitions, probabilities of state transitions were computed and then used to construct the edges of the Markov chain model with the formula [10]:

$$\begin{aligned} & \text{Probability of state } S_i \text{ transitioning to state } S_j \\ &= \frac{\text{number of transitions from state } S_i \text{ to state } S_j}{\text{number of transitions from state } S_i} \end{aligned} \tag{3.3.1}$$

To validate the k-means derived transition probabilities and stationary distribution, a hidden Markov model (HMM) was trained in order to learn the optimal state transitions probabilities based on the survey data. Given that HMMs have observed variables and hidden states [13][15], the observed variables and hidden states correspond to the survey data and coping profiles, respectively. The HMM's input are paired sequences of data and the number of hidden states [16]. For the paired sequences, the unlabelled survey responses were divided into paired sequence of responses from each respondent. Five hidden states were designated as the number of hidden states to rediscover the same coping profiles identified by the k-means clustering model. Learning of the state transition probabilities is done through the

Baum-Welch Algorithm which is an iterative method of updating the model's parameters until the model converges to a local minimum or maximum [13][17]. The HMM composed of the state transition probabilities A , emission probabilities B , and initial state probabilities π are updated using the following formula:

$$\bar{\pi}_i = \text{expected number of times in state } S_i \text{ at time } (t = 1) \quad (3.3.2)$$

$$\bar{a}_{ij} = \frac{\text{expected number of transitions from state } S_i \text{ to state } S_j}{\text{expected number of transitions from state } S_i} \quad (3.3.3)$$

$$\bar{b}_j(k) = \frac{\text{expected number of times in state } j \text{ and observing symbol } v_k}{\text{expected number of times in state } j} \quad (3.3.4)$$

The k-means derived Markov Chain and HMM were simulated using the Monte-Carlo technique to produce a stationary distribution in the long run which represents the coping profile distribution in equilibrium [10][11]. The resulting stationary distributions from both Markov models were compared.

Evaluation of the HMM is based on how similar the HMM labeled results are to the k-means clustering labeled results, which was assessed through classification accuracy and the area under the receiver operating characteristic (ROC) curve. To account for data imbalance, stratified shuffle split with five iterations was performed. Aside from the classification metrics, the HMM was used to generate samples which were classified by the k-means clustering model to assess the HMM's performance on producing samples that can be accurately predicted by the k-means clustering model. Visual comparisons were also done with the radio plots of the 10 defining features of the HMM and the k-means clustering.

3.4 Statistical Tests

Differences or similarities accompanying coping profile transitions across different surveys were identified through nonparametric statistical tests. Nonparametric tests were selected to relax the normality assumption given the significant number of features present in the survey and the difference in how the features were measured. For all tests, a significance level of $\alpha = 0.05$ was set.

The Kruskal-Wallis statistical test is used to compare the medians of different populations [20]. In this case, Kruskal-Wallis tests were applied on the coping profiles to compare the medians of the defining features: UCLA loneliness score, Subjective Happiness Scale (SHS) score, Posttraumatic Growth Inventory-Expanded (PTGI-X) score, Kessler Psychological Distress Scale (K6) score, amount of exercise, deterioration in social interaction, deterioration in economy, optimism, trust in government, and continuous practice of preventive actions against the pandemic. This was done to check for significant changes or similarities within the same coping profiles but across different time periods.

The Wilcoxon signed-rank statistical test can be used to check for significant differences between paired data (e.g., before and after data) [20]. For significant differences with respect to categorical variables, the McNemar statistical test can be applied on paired data [21]. Wilcoxon signed-rank and McNemar tests were performed on numerical and categorical features respectively to check for any significant changes in the features after a coping profile transition. Note that one-tailed Wilcoxon signed-rank tests were conducted to check if the change in the feature was a significant increase or decrease over time.

Chapter 4

Results

In this chapter, the resulting data after data preprocessing is presented. Clusters identified in different time points are compared via inspection of radio plots of the 10 defining features which the k-means clustering model utilizes to determine a respondent's coping profile. The k-means derived Markov chain is compared to an HMM learned Markov chain through classification metrics. Both stationary distributions derived from the Markov chains are also analyzed. Statistical test results are shown to enumerate feature changes related to coping profile changes.

4.1 Data Preprocessing

Following the data preprocessing steps from the first phase of the project results to the filtered data with coping profiles (N2=15493; N3=8899; N4=4888; N5=13415). Each survey consists of the the following features:

1. Sociodemographic - age, sex, civil status, presence of children in the household, number of household members, household income, personal income, job group, year level if respondent is a student, if someone in the family or the respondent is a medical professional, current or past treatment of physical illness, current

- or past treatment of mental illness,
2. COVID-related - current or past treatment of COVID, interaction deterioration, economy deterioration, perception of COVID related supplies, perception of work/study changes,
 3. Mental health - K6, PHQ9, GAD7, SSS8, PTGIX, MAIA, SHS, UCLA, LSNS6, frustration due to lifestyle changes, COVID-19 anxiety, COVID-19 insomnia,
 4. Coping behaviors - exercise, healthy diet, activities that inspire, offline and online interaction, engaged in continuous prevention of COVID-19, engaged in prevention of COVID-19 to prevent transmission to family members or others, engaged in prevention of COVID-19 to avoid social criticism and pressure, understanding of the importance of COVID-19 prevention, trust in the government, trust in social media, vaccination willingness, optimism, healthy sleep

The samples were further processed to paired sequences of data since coping profile transitions require a before and after state. This limits the filtered data to only responses that have a succeeding response in the next time point. Consequently, the original data was reduced to Survey 2 to 3 (N=8370), Survey 3 to 4 (N=4031), and Survey 4 to 5 (N=4131) transitions, or in total 33064 responses. Distributions of sociodemographic features using the fully processed data are presented in Figures A.1 to A.4.

4.2 K-means Clustering

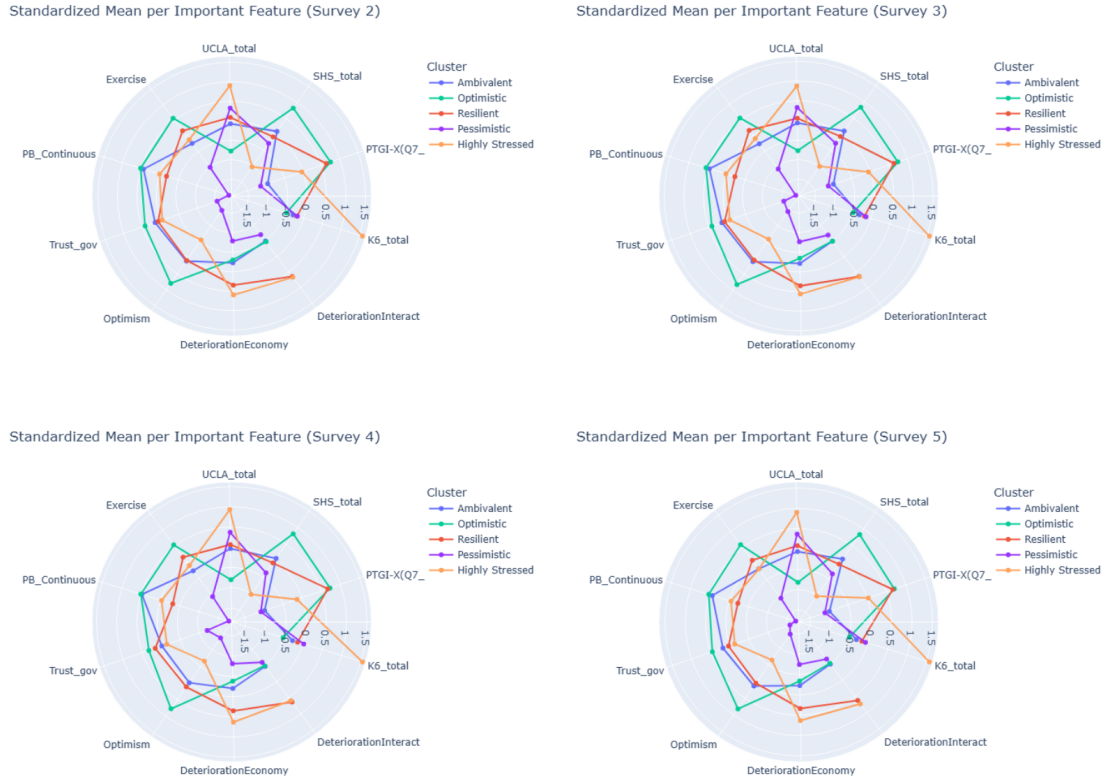


Figure 4.1: Radio plots of standardized mean per important feature across all surveys

Radio plots of the standardized mean for each coping profile per survey clustered using the k-means algorithm are provided in Figure 4.1. Observe that the structures of the 10 defining features are consistent with the trained data’s radio plot (Survey 5). Cluster distributions across the surveys are displayed in Figure A.5. The cluster distributions suggest that the coping profiles follow a consistent distribution: Resilient ($\approx 30\%$), Ambivalent ($\approx 25\%$), Optimistic ($\approx 22\%$), Highly Stressed ($\approx 15\%$), and Pessimistic ($\approx 8\%$).

4.3 Markov Models

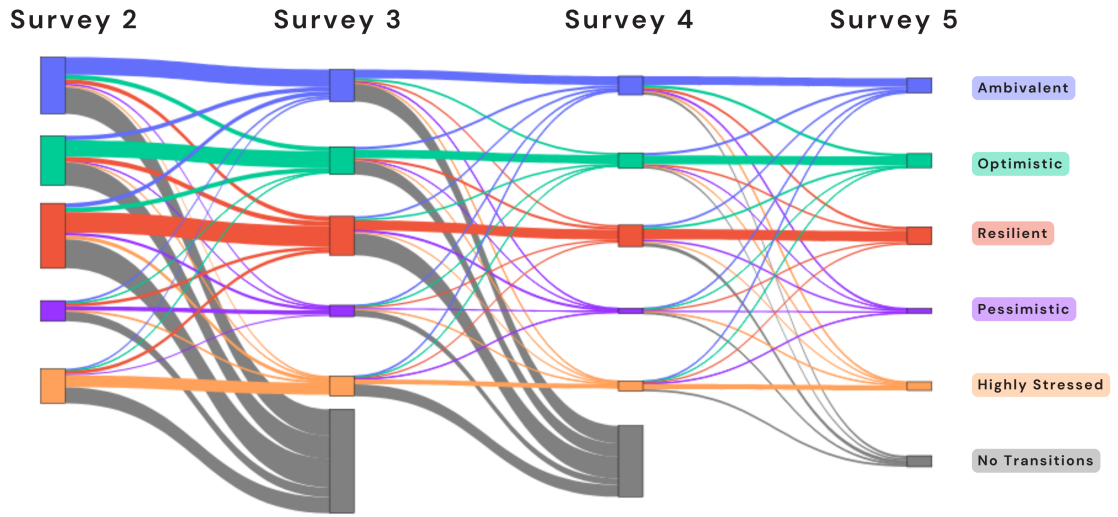


Figure 4.2: Sankey diagram of coping profile transitions across different surveys

Following the prediction of coping profiles across different surveys, the coping profile transitions are visualized through a Sankey diagram in Figure 4.2. Samples who were not able to respond to the next chronological survey were omitted.

The frequencies for each coping profile transition are listed in Table 4.1a. These frequencies are also converted to transition probabilities in Table 4.1b. The transition probabilities are used to construct the Markov chain which is composed of the graphs in Figure 4.3. Majority of the transition probabilities go back to the same coping profile except for the Pessimistic profile wherein there is only a 37% probability of going back to the Pessimistic profile. Other major transition probabilities across all profiles point towards the Ambivalent, Optimistic or Resilient coping profiles.

Table 4.1: (a) Coping profile transition frequencies and (b) transition probabilities

To						
From		Ambivalent	Optimistic	Resilient	Pessimistic	Highly Stressed
	Ambivalent	2409	639	621	256	226
	Optimistic	657	2338	628	51	67
	Resilient	574	572	2834	398	465
	Pessimistic	279	51	418	519	128
	Highly Stressed	224	60	435	159	1524

To						
From		Ambivalent	Optimistic	Resilient	Pessimistic	Highly Stressed
	Ambivalent	0.58	0.15	0.15	0.06	0.06
	Optimistic	0.18	0.62	0.17	0.01	0.02
	Resilient	0.12	0.12	0.58	0.08	0.10
	Pessimistic	0.20	0.04	0.30	0.37	0.09
	Highly Stressed	0.09	0.02	0.18	0.07	0.63

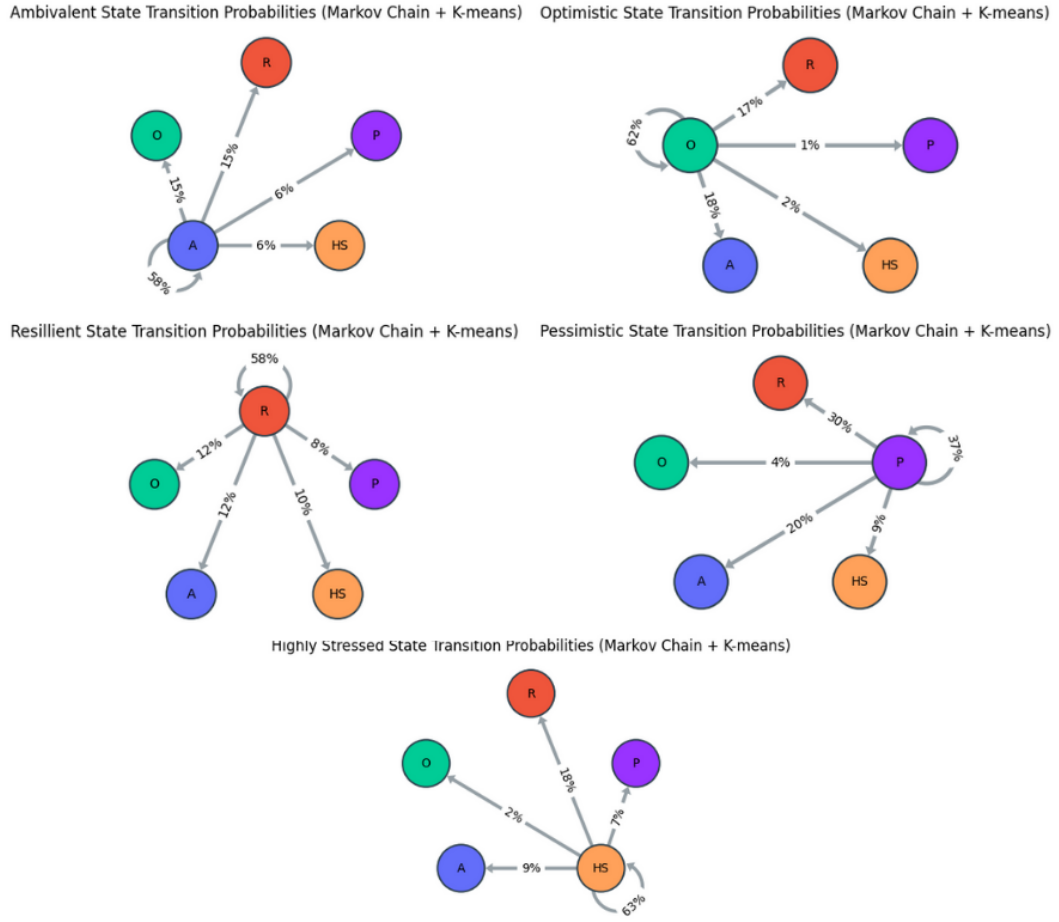


Figure 4.3: Markov chain constructed with k-means clustering

Moving on to the training of the HMM on the paired sequences of data, the cross-validation results and the test accuracy of the HMM based on k-means labeled data are found in Table 4.2. On the other hand, accuracy of the k-means clustering model on different sets of HMM generated samples are shown in Table 4.3. Note that the trained HMM has a silhouette score of 0.0776 which indicates poor clustering performance given that some of the clusters have overlaps. Despite that, the radio plots produced by the HMM resemble the radio plots produced by the k-means clustering model as can be observed in Figure 4.4. The accurate results of the HMM on k-means

labelled data combined with the highly similar radio plot of the 10 defining features imply that the same coping profiles were identified by the HMM.

Table 4.2: General classification metrics

Iteration	Accuracy	ROC
Cross-val Split 0	88%	0.98
Cross-val Split 1	87%	0.98
Cross-val Split 2	87%	0.98
Cross-val Split 3	86%	0.98
Cross-val Split 4	87%	0.98
Test Set	87%	0.98

Table 4.3: Accuracy of k-means prediction on HMM generated samples

Number of generated samples	Number of correctly predicted samples using k-means	Accuracy
10,000	9,998	99%
30,000	29,997	99%
50,000	49,995	99%

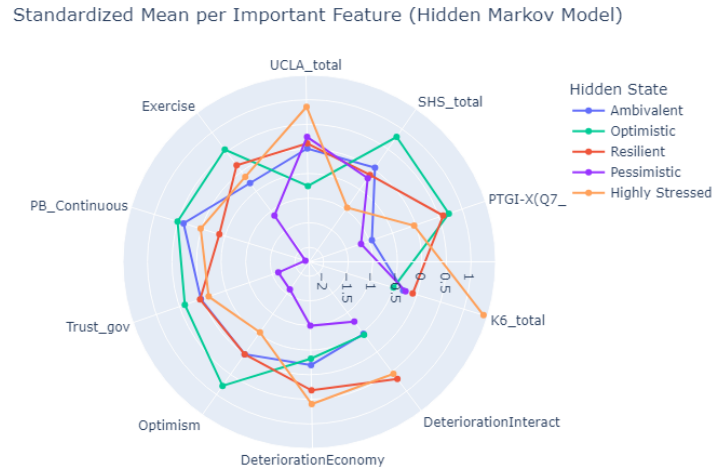


Figure 4.4: Radio plot of standardized means per important feature (HMM)

Table 4.4: Learned transition probability matrix (HMM)

		To				
From		Ambivalent	Optimistic	Resilient	Pessimistic	Highly Stressed
	Ambivalent	0.65	0.14	0.12	0.03	0.06
	Optimistic	0.16	0.69	0.11	0.01	0.03
	Resilient	0.11	0.10	0.63	0.07	0.09
	Pessimistic	0.22	0.05	0.32	0.31	0.10
	Highly Stressed	0.09	0.04	0.14	0.05	0.68

The HMM learned the optimal transition probability matrix in Table 4.4 given the data from the training process. This matrix corresponds to the transition graphs present in the HMM learned Markov chain as seen in Figure 4.5. Similar to the k-means derived Markov chain, majority of the coping profile transitions are towards the same coping profile except for the Pessimistic profile where majority of its profile transitions are evenly split between the Pessimistic (31%) and Resilient (32%) profiles. The next major transitions go towards either the Ambivalent, Optimistic or Resilient coping profiles.

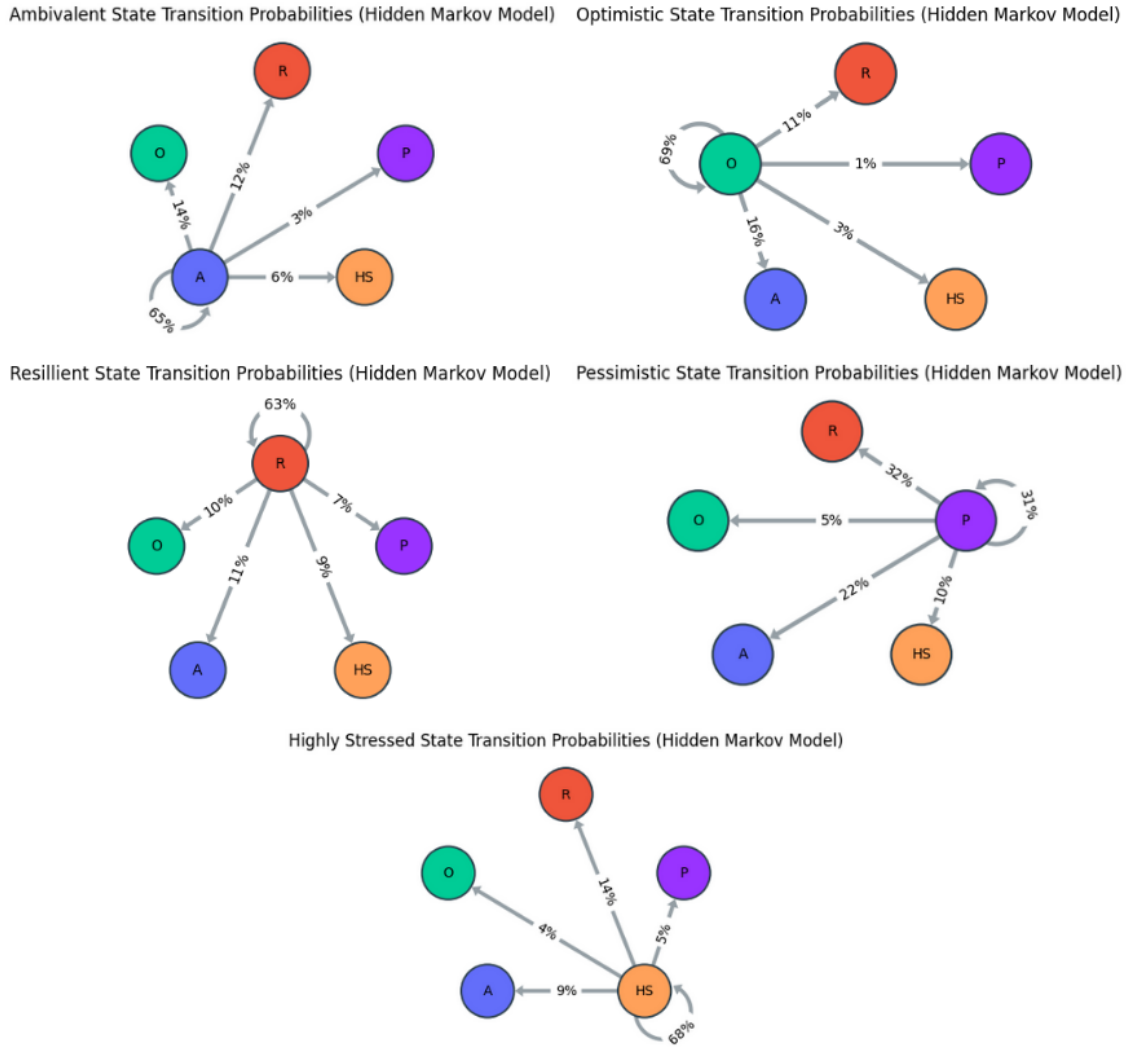


Figure 4.5: Markov chain learned by the HMM

With the two Markov chains from the transition counts and learned transition probability matrix, stationary distributions were simulated using the Monte-Carlo simulation technique. The resulting stationary distributions are illustrated in Figure 4.6 and the values are included in Table 4.5. The values indicate that the distribution presented in Figure A.5 will remain consistent even as the coping profiles reach equilibrium, with majority of the coping profiles being within the Ambivalent, Optimistic

and Resilient profiles.

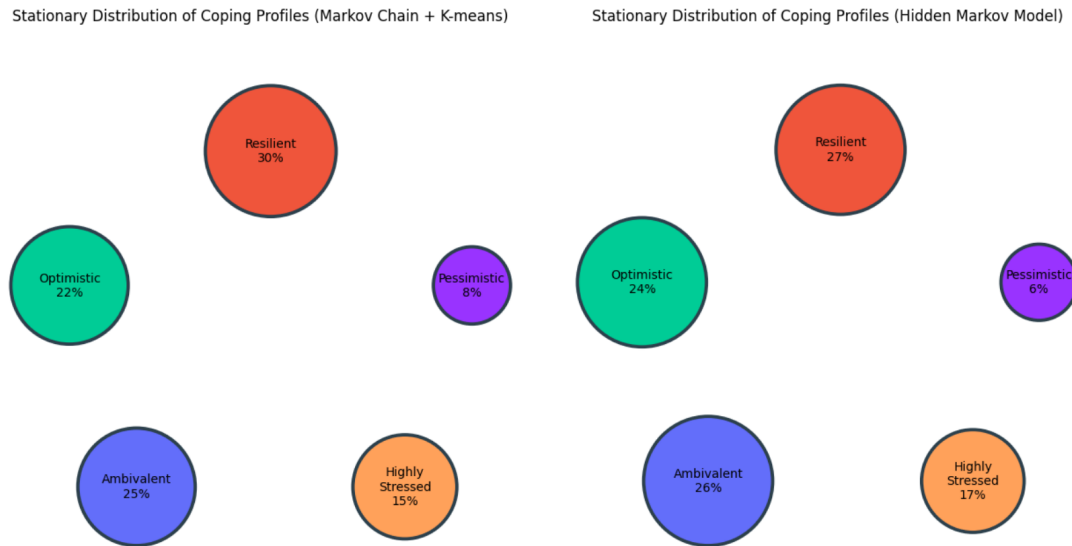


Figure 4.6: Simulated stationary distributions of k-means Markov chain and HMM

Table 4.5: Predicted coping profile distributions of the population in the long run

	Ambivalent	Optimistic	Resilient	Pessimistic	Highly Stressed
k-means Markov chain	0.25	0.22	0.30	0.08	0.15
HMM	0.26	0.24	0.27	0.06	0.17

4.4 Statistical Tests

Box plots for each cluster over time were created. An example of said box plots is presented in Figure B.5 which shows the averages of features within the Ambivalent cluster across the different surveys. From the plots, it can be seen that there are differences in the 10 defining features within the same coping profile but across different time periods. The significant differences present in the visual results are confirmed through the p-values of the Kruskal-Wallis test performed on the important features of the same clusters but across different time points. The results from said test are summarized in Table 4.6. Enumerated below are the features of coping profiles that did not exhibit significant changes throughout the pandemic.

1. Ambivalent - UCLA, Economy Deterioration
2. Optimistic - UCLA, Interaction Deterioration, Continuous Prevention
3. Resilient - PTGIX, Interaction Deterioration, Economy Deterioration
4. Pessimistic - UCLA, SHS
5. Highly Stressed - UCLA, K6, SHS, Continuous Prevention, Exercise

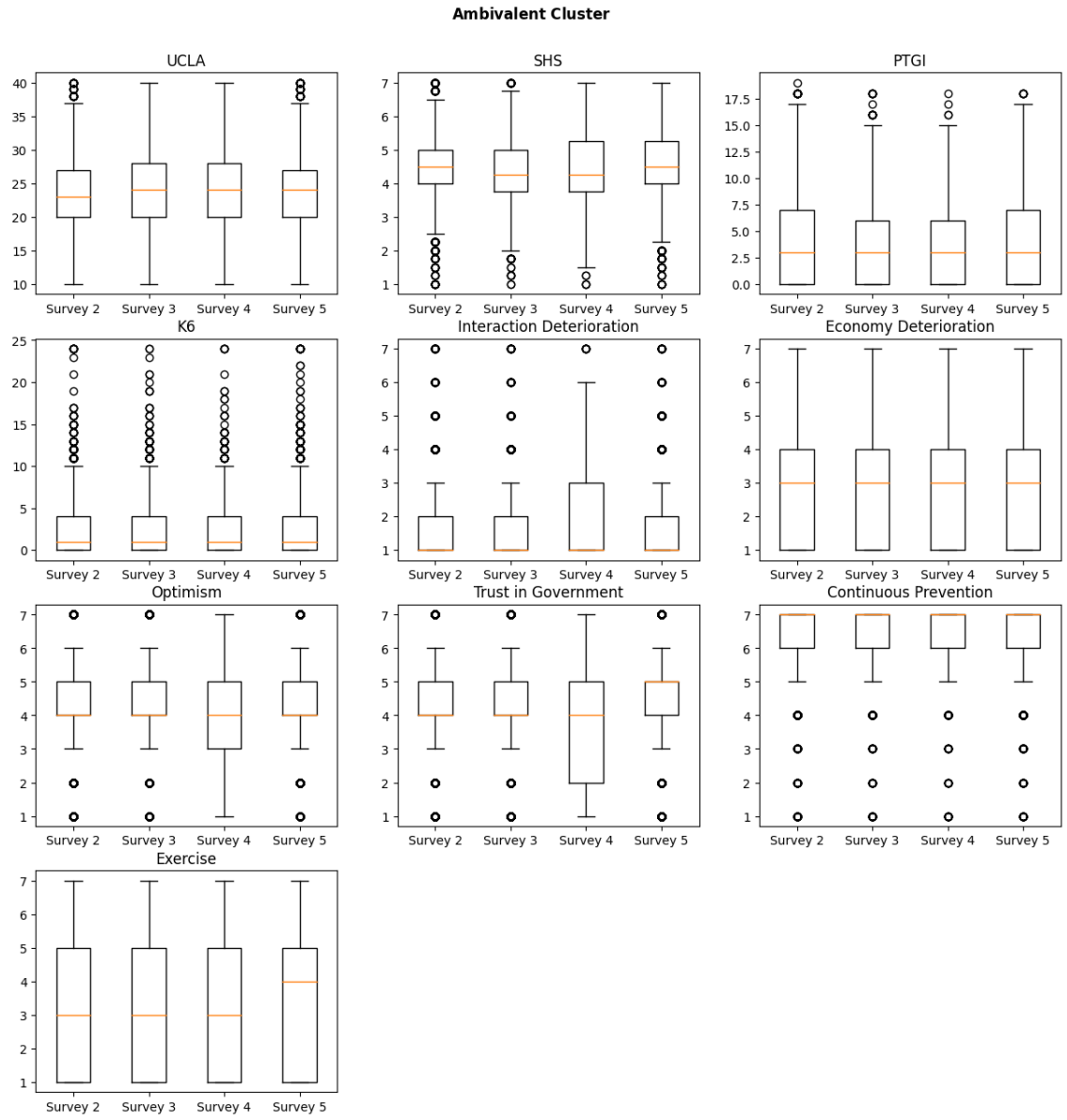


Figure 4.7: Box plots of the Ambivalent Cluster's features

Table 4.6: P-values of Kruskal-Wallis test on the 10 defining features within same clusters over time

Feature	Ambivalent	Optimistic	Resilient	Pessimistic	Highly Stressed
UCLA	0.12	0.73	0.03	0.13	0.42
SHS	0.00*	0.00*	0.00*	0.39	0.85
PTGI-X	0.00*	0.00*	0.99	0.00*	0.00*
K6	0.02*	0.04*	0.00*	0.00*	0.18
Exercise	0.00	0.00	0.04*	0.00	0.43
Interaction Deterioration	0.00*	0.37	0.11	0.00*	0.00*
Economy Deterioration	0.09	0.00*	0.16	0.00*	0.01*
Optimism	0.00*	0.00	0.00*	0.00*	0.00*
Trust in Government	0.00*	0.00*	0.00*	0.00*	0.00*
Continuous Prevention	0.00*	0.08	0.00*	0.00	0.34

* $p < 0.05$

A portion of numerical features' mean values in all coping profiles are displayed in Table 4.7. There are disparities between the mean values of the different coping profiles which imply that a coping profile transition may result to a significant change with respect to a certain feature. These significant changes are confirmed in the results from the McNemar and Wilcoxon signed-rank tests on samples that experienced coping profile transitions. A part of the results from said tests on the Ambivalent coping profile are presented in Table 4.8.

Table 4.7: Mean values of numerical features across all clusters

Feature	Ambivalent	Optimistic	Resilient	Pessimistic	Highly Stressed
Household income*	3.52	4.02	3.67	3.32	3.11
Personal income*	2.33	2.61	2.60	2.52	2.12
K6	2.45	1.46	3.12	3.42	12.32
PTGIX	3.96	18.38	17.69	2.71	11.97
SHS	4.44	5.25	4.27	4.00	3.23
UCLA	23.55	19.33	24.36	25.90	29.13
Exercise	3.31	4.81	4.01	1.74	3.49
Healthy diet	4.06	5.20	4.26	1.94	3.85
Favorite activity	3.34	4.76	4.02	1.81	3.40
Interaction offline	3.23	4.24	3.78	1.75	3.16
Interaction online	1.99	3.18	3.30	1.41	2.64
PB Continuous	6.26	6.39	5.13	2.46	5.48
Trust gov	4.29	4.73	4.18	1.74	3.95
Vaccination will	5.02	5.37	4.51	1.90	4.42
Optimism	4.32	5.42	4.28	1.93	3.24
Economy deterioration	3.12	2.93	4.10	2.17	4.54
Interaction deterioration	1.85	1.85	3.61	1.57	3.68
COVID anxiety	2.91	3.03	4.12	1.61	4.56
Difficulty living	1.62	1.83	3.68	1.45	3.55

* Income Scale: 1 = less than 1M yen, 2 = 2M to 4M yen, 3 = 4M to 6M yen

Table 4.8: McNemar and Wilcoxon-signed rank results on cluster transitions with Ambivalent cluster as source

Feature	Ambivalent	Optimistic	Resilient	Pessimistic	Highly Stressed
Household income					
Personal income		increased			
K6		decreased			increased
PTGIX	decreased	increased	increased		increased
SHS		increased		decreased	decreased
UCLA		decreased		increased	increased
Exercise		increased	increased	decreased	
Healthy diet		increased		decreased	increased
Favorite activity	increased	increased	increased	decreased	
Interaction offline	increased	increased	increased	decreased	increased
Interaction online		increased	increased	decreased	increased
PB Continuous			decreased	decreased	decreased
Trust gov		increased		decreased	
Vaccination will	increased	increased	decreased	decreased	
Optimism		increased		decreased	decreased
Economy deterioration			increased	decreased	increased
Interaction deterioration	increased	increased	increased		increased
Covid Anxiety	decreased		increased	decreased	increased
Difficulty living	decreased		increased	decreased	increased

Chapter 5

Discussion

Real life implications of the resulting Markov models and statistical tests are discussed in this section. Results from studies made during the COVID-19 pandemic were compared with the findings.

5.1 Coping Profile Dynamics

The results from the Markov chain modeling show that the learned HMM's Markov chain and the k-means derived Markov chain have small differences with regards to the transition probabilities. This small difference in the transition probabilities may be due to the training of the HMM which utilized paired sequences of observed data and the HMM using both the means and covariance of the data to determine the coping profile of a sample unlike the k-means clustering model using only the means of the data. However, despite the differences, the HMM model was able to distinguish the same coping profiles identified by the k-means clustering model visualized in Figure 4.4 thus supporting the model and findings from the first phase of this project.

From the learned transition probabilities in the HMM, it can be concluded that samples in the Pessimistic profile have decent probabilities of transitioning to the

Resilient (32%) or Ambivalent (22%), in contrast with the other profiles which would most likely transition to the same profile (greater than 60%). As for the stationary distributions, the majority of the population is predicted to be in the Optimistic, Resilient and Ambivalent profiles. Movement towards the more neutral and positive profiles are indicators of the public adapting to the pandemic and the introduction of vaccines. These are supported by the statistical tests which found that in the majority of the transitions, there was a decrease in perceived shortage of supplies related to the prevention of COVID (masks, thermometers) and anxiety brought by news about the pandemic, while there was an increase in vaccination willingness and offline interaction.

5.2 Related Factors to Coping Profile Transitions

Analyzing further the results from the statistical tests, the consistency of the UCLA scores within the same profiles over time are supported by Sugaya et al's [3] study wherein their results found no significant improvements in the loneliness levels between two time points: 1st and 2nd mild lockdown period in Japan. The result in this study is able to extend Sugaya's results up to the 3rd, 4th and 5th mild lockdown periods in Japan suggesting that the loneliness issue goes beyond the initial stress caused by the pandemic. However, this consistency in loneliness is only found within the Ambivalent, Optimistic, Pessimistic and Highly Stressed profiles. More exploration may be done on an individual profile for future studies.

All profiles transitioning to the Highly Stressed profile are related to increase in frustrations due to lifestyle changes, worsening economic situation, and anxiety caused by news about the pandemic. Despite numerous studies suggesting low income

[22][23] as a risk factor to mental health decline, there was no significant decrease in income with respect to transitions to the Highly Stressed profile. Relating this to the worsening economic situation of the Highly Stressed profile, additional expenses (e.g. supplies for COVID prevention), weakening of Japanese yen as broadcasted in media, and reintroduction of travel expenses may have been some of the primary stressors of said profile. This study reinforces the relationship between financial difficulties and mental health problems as also concluded by other studies [24][25] studies. Common changes in all profile transitions to the Optimistic profile were characterized by an increase in exercise, social interactions, healthy diet, pursuing hobbies, healthy sleep, and optimism. The positive effects of these coping behaviors are also mentioned in multiple findings [26][5][27][4][28] wherein active coping such as exercise, social interactions and optimism were associated with lower levels of depressive symptoms. Moreover, profile transitions from Resilient and Ambivalent to the Optimistic profile was related to an increase in income, which if combined with other studies [22][23][3] mentioning that reduced income was a risk factor to mental health, suggests a positive correlation between income and mental health during the pandemic.

5.3 Limitations

It was assumed that the model from the k-means clustering model is the correct model given that it was validated during the first phase of this project. Consequently, the HMM's training objective was to identify the same coping profiles as the k-means clustering model.

Individual and aggregated PTG scores were included during preprocessing which are correlated. The HMM may take the correlation of the data into account, however,

the k-means clustering model is limited to the means of the data.

Coping behaviors considered in this study are mostly positive. This limitation may have led to the positive outlook in the results related to coping behaviors. Future studies may look into other coping mechanisms which can be detrimental to one's mental health.

Chapter 6

Summary and Conclusion

The dynamics of coping profiles throughout the duration of the COVID-19 pandemic was modeled using Markov Models. Given how the Japanese government handled the pandemic in the major prefectures, it is expected that the majority of the Japanese population would transition to neutral and positive profiles: Ambivalent, Resilient and Optimistic. This movement is related to the public adapting to the pandemic and the introduction of vaccines as indicated by a decrease in perceived shortage of supplies related to the prevention of COVID (masks, thermometers) and anxiety brought by news about the pandemic, and an increase in vaccination willingness and offline interaction.

Despite adjusting to the pandemic, loneliness remained an issue in most of the coping profiles. It may be inferred that loneliness goes beyond the pandemic within the Japanese population. Worsening financial struggles were consistent in transitions to the Highly Stressed profile. As for transitions to the Optimistic profile, the common factors were an increase in income, and increase in frequency of the coping activities: exercise, healthy diet, pursuing hobbies, healthy sleep, and optimism.

As a practical application of these results, government institutions may look into

the movement of coping profiles as a metric of the public's adaptation towards a pandemic. Low income groups may be targeted by mental health intervention programs. Furthermore, financial assistance programs should be implemented by the government and employers to alleviate the additional expenses caused by a public crisis which are significant stressors. For individuals, mental health states may improve through the following activities: exercising, consuming a healthy diet, pursuing activities that inspire, socially interacting with family and friends, sleeping sufficiently and having a positive outlook on the future.

The observed coping profile dynamics, mental health risk factors, and recommended coping behaviors can be explored further by mental health studies focusing on the Philippine population. The Philippine government may also plan around low income groups and the recommended coping behaviors for mental health interventions during a pandemic.

Appendix A

Exploratory Data Analysis

A.1 Sociodemographic Features

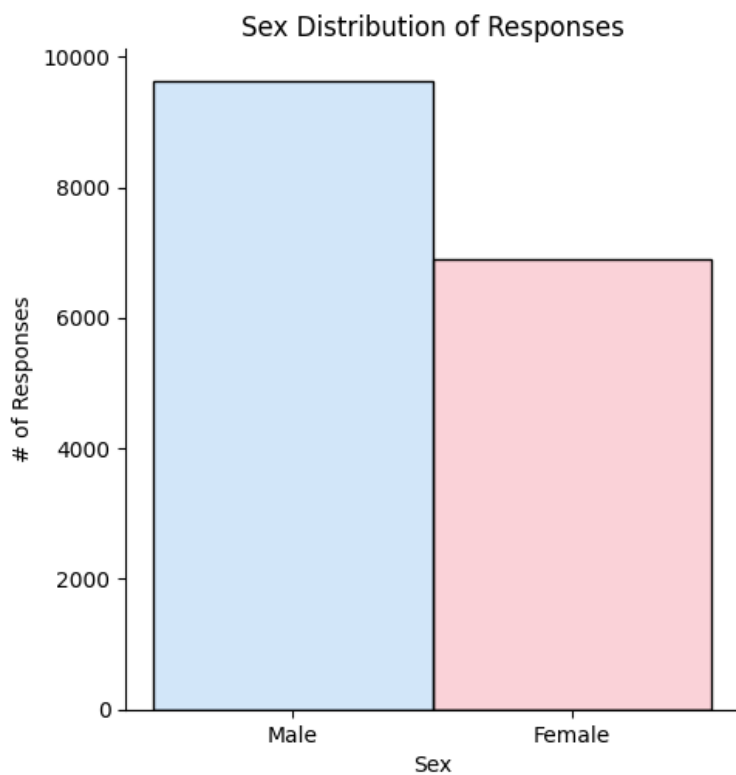


Figure A.1: Distribution plot of sex across all surveys

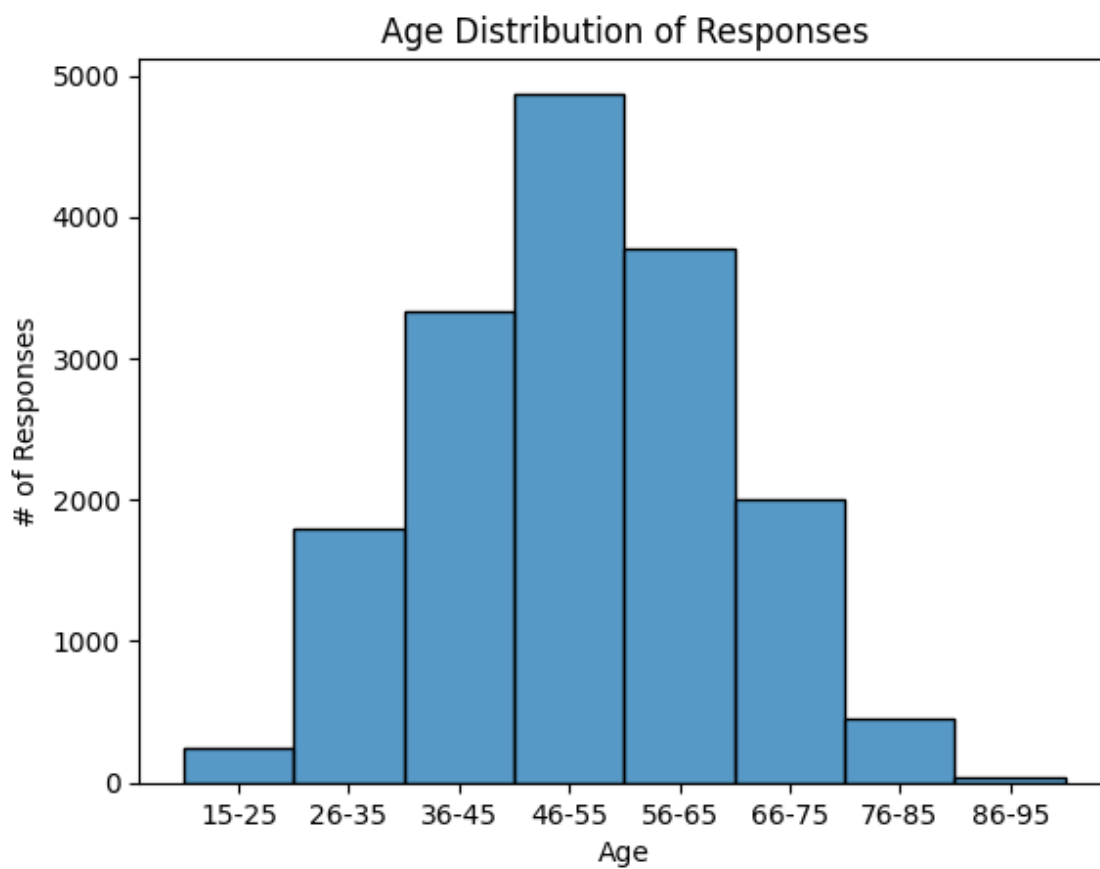


Figure A.2: Distribution plot of age across all surveys

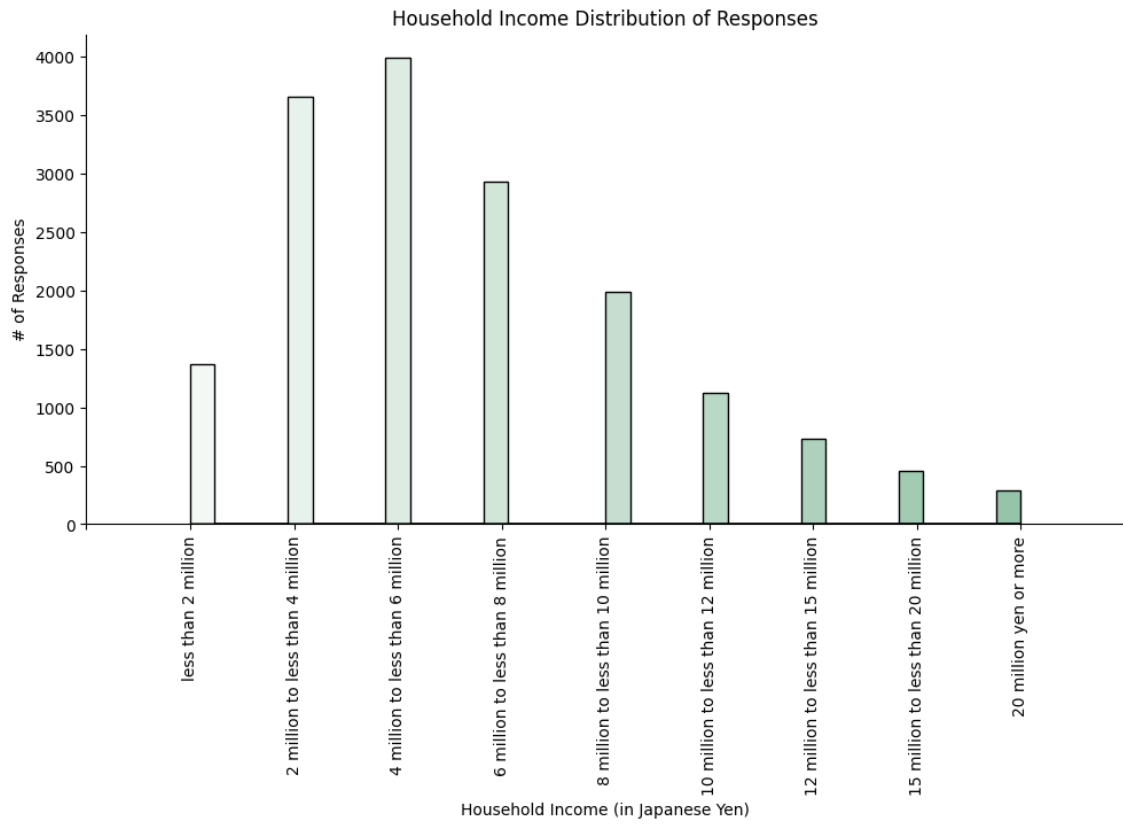


Figure A.3: Distribution plot of household income across all surveys

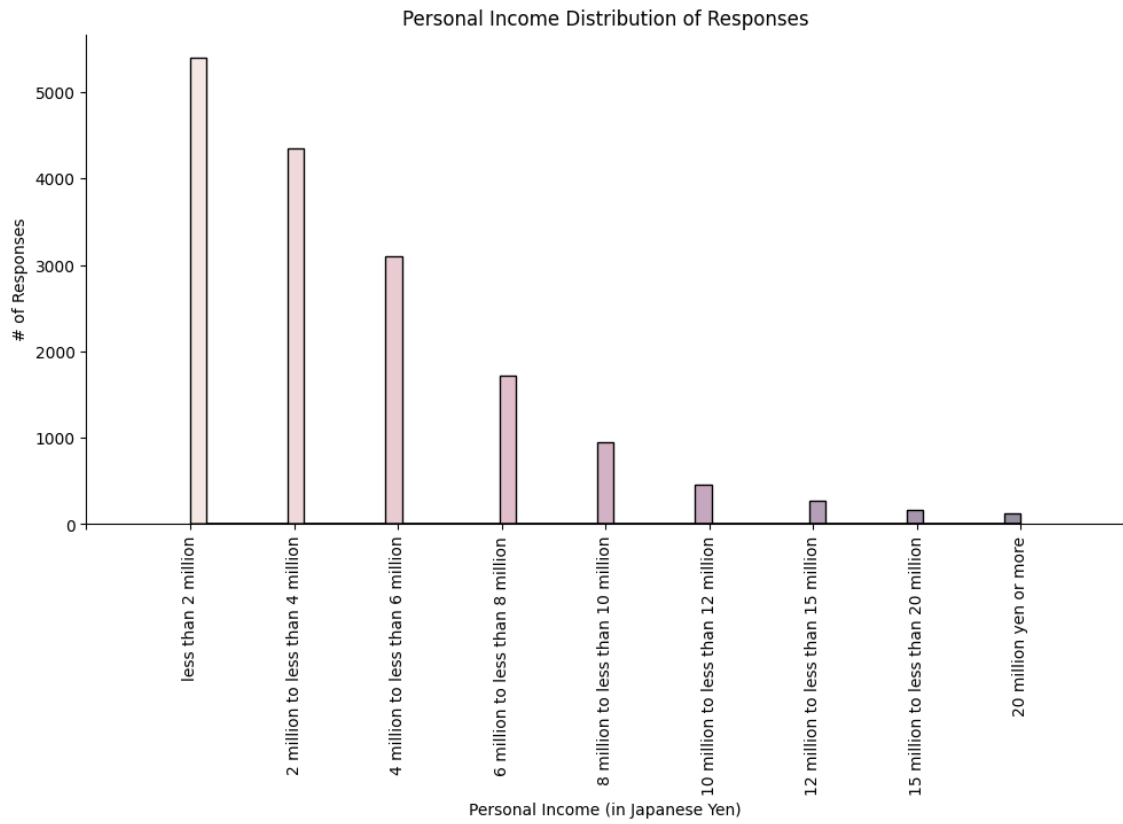


Figure A.4: Distribution plot of personal income across all surveys

A.2 Coping Profile Distribution

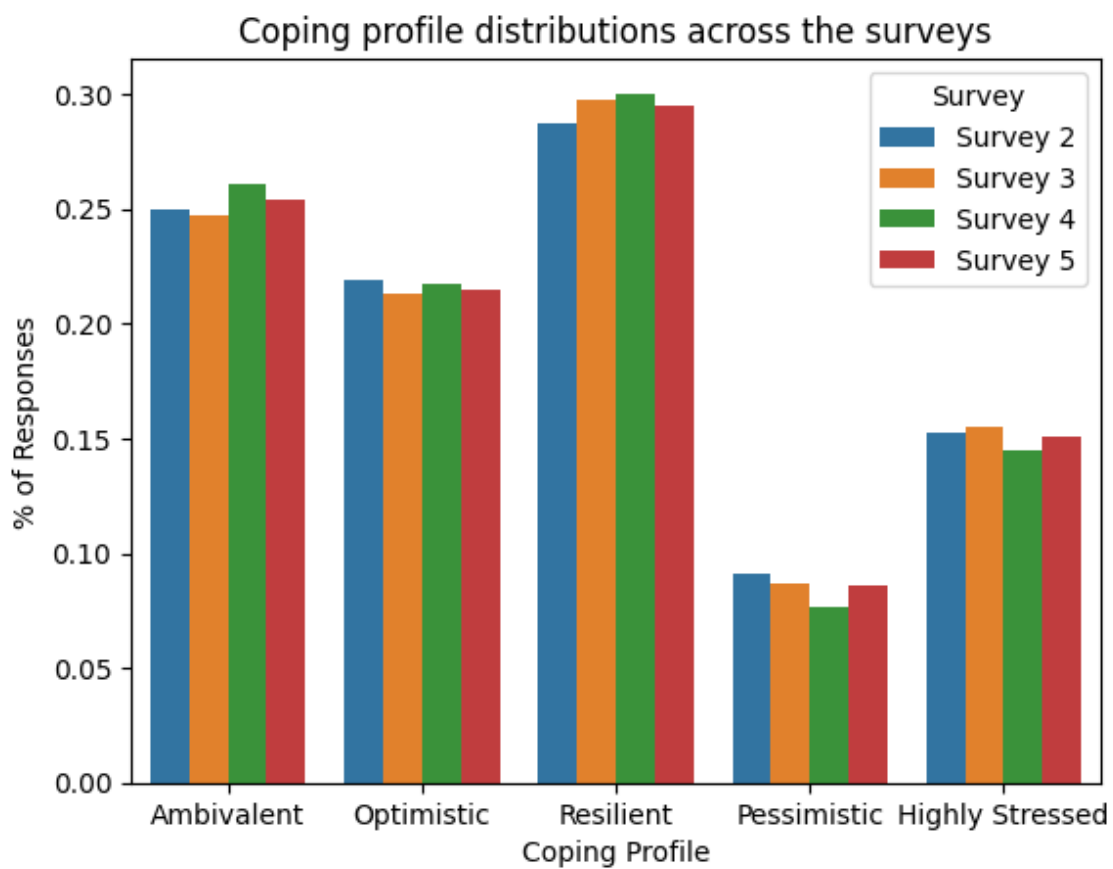


Figure A.5: Distribution plot of coping profiles across all surveys

Appendix B

Statistical Test Results

B.1 Box Plots of the Defining Features

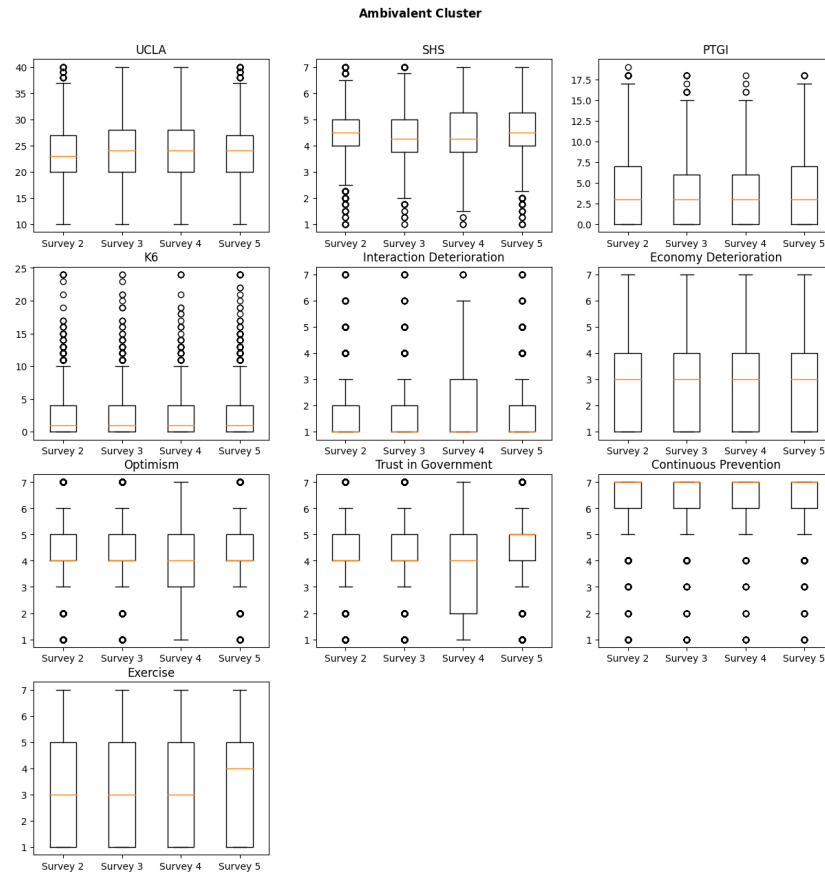


Figure B.1: Box plots of the Ambivalent Cluster's features

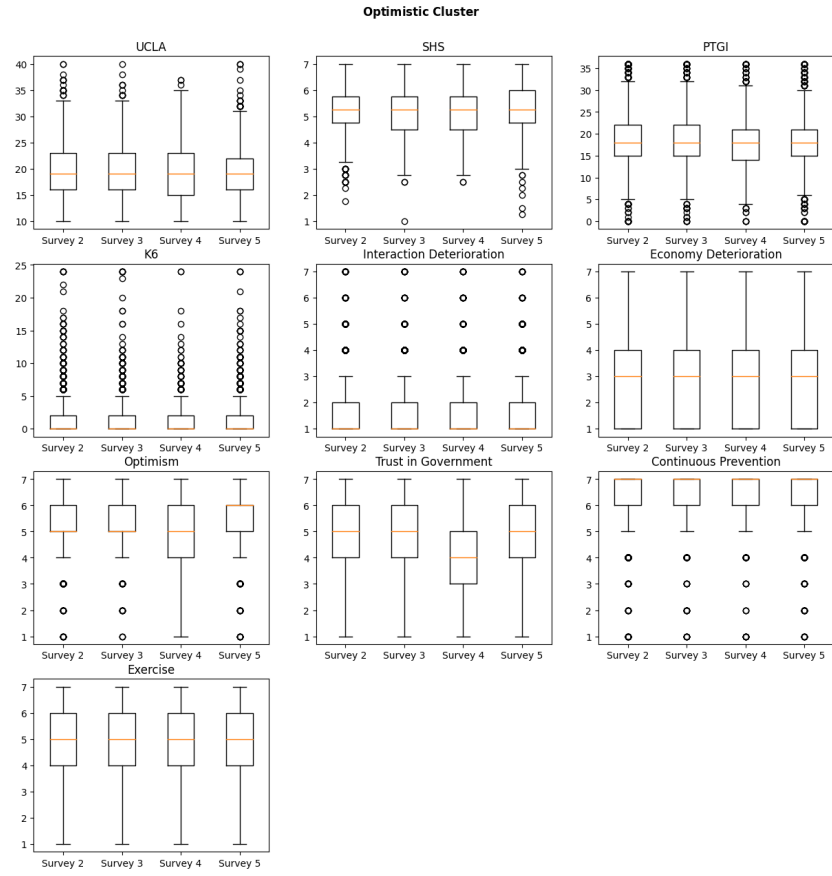


Figure B.2: Box plots of the Optimistic Cluster's features

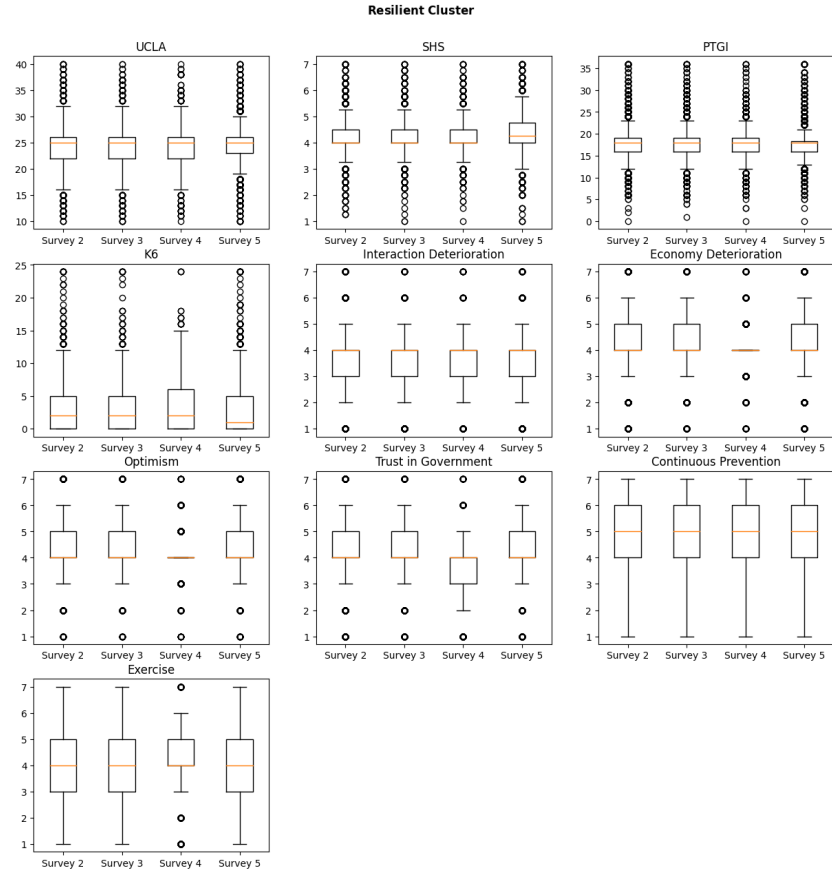


Figure B.3: Box plots of the Resilient Cluster's features

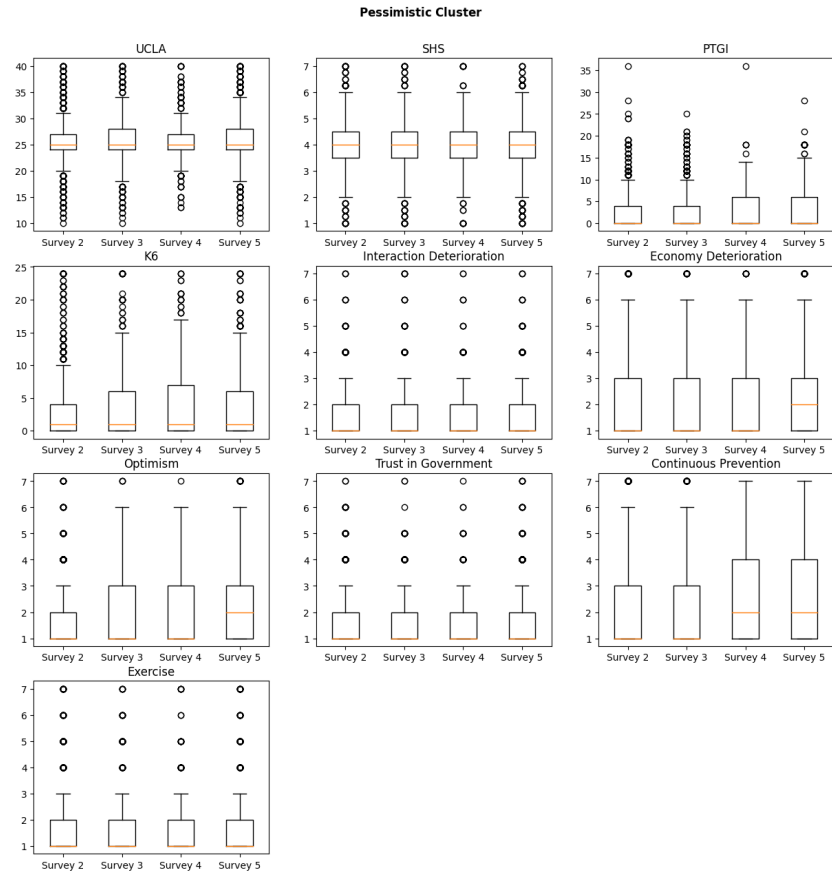


Figure B.4: Box plots of the Pessimistic Cluster's features

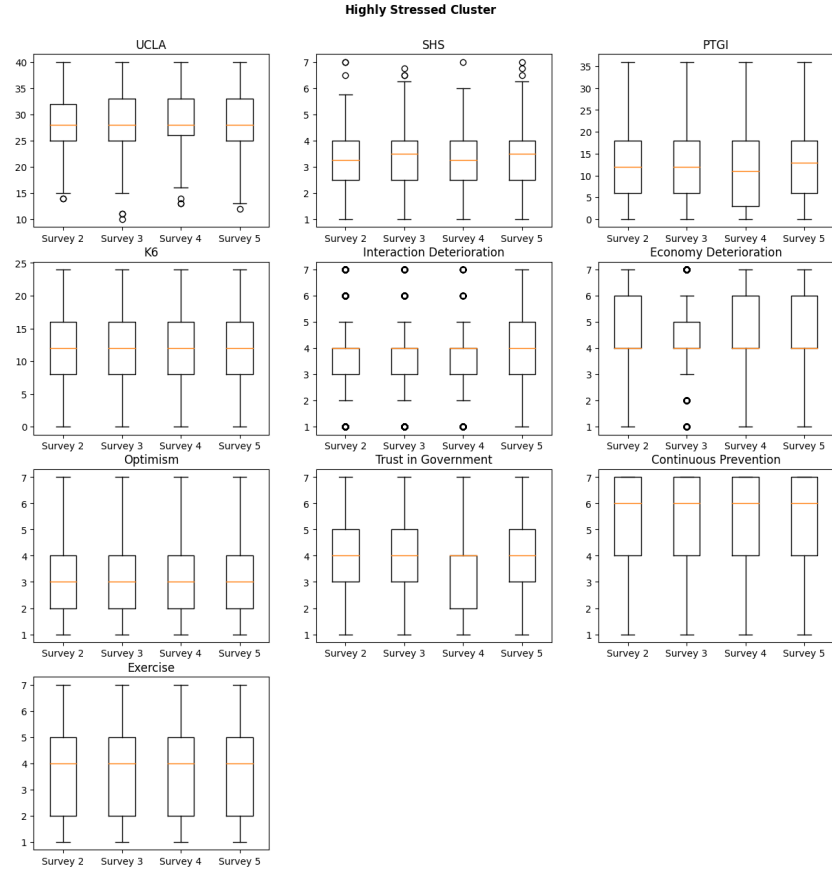


Figure B.5: Box plots of the Highly Stressed Cluster's features

B.2 McNemar and Wilcoxon Signed-rank Tests Tables

Table B.1: Mean values of numerical features across all clusters

Feature	Ambivalent	Optimistic	Resilient	Pessimistic	Highly Stressed
Household income	3.52	4.02	3.67	3.32	3.11
Personal Income	2.33	2.61	2.60	2.52	2.12
Number fam	1.82	1.94	2.19	1.93	1.96
School grade	0.01	0.02	0.02	0.02	0.04

Years enrolled	0.01	0.03	0.02	0.03	0.05
K6	2.45	1.46	3.12	3.42	12.32
PHQ9	2.43	1.35	2.51	3.24	13.74
GAD7	1.58	0.97	1.89	2.01	10.62
SSS8	3.99	3.15	3.98	3.46	13.50
PTGIX	3.96	18.38	17.69	2.71	11.97
MAIA 1	1.22	1.67	1.48	0.78	2.55
MAIA 2	3.64	3.63	3.52	4.16	2.24
MAIA 3	2.78	2.96	2.74	3.01	2.04
MAIA 4	2.11	2.97	1.64	0.84	2.26
MAIA 5	1.94	3.02	1.69	0.81	2.33
MAIA 6	2.09	3.18	1.75	0.85	2.12
MAIA 7	1.76	2.99	1.68	0.78	2.08
MAIA 8	2.10	3.29	1.78	0.88	1.97
SHS	4.44	5.25	4.27	4.00	3.23
UCLA	23.55	19.33	24.36	25.90	29.13
LSNS6	9.28	13.34	9.50	6.29	6.79
Frustration	2.47	2.35	3.99	1.76	4.67
Exercise	3.31	4.81	4.01	1.74	3.49
Healthy diet	4.06	5.20	4.26	1.94	3.85
Favorite activity	3.34	4.76	4.02	1.81	3.40
Interaction offline	3.23	4.24	3.78	1.75	3.16
Interaction online	1.99	3.18	3.30	1.41	2.64
PB Continuous	6.26	6.39	5.13	2.46	5.48
PB Altruistic	6.06	6.26	5.04	2.26	5.36

PB Avoidant	5.34	5.52	4.80	2.28	5.10
Trust gov	4.29	4.73	4.18	1.74	3.95
Trust social media	3.06	3.80	3.85	1.57	3.51
Vaccination will	5.02	5.37	4.51	1.90	4.42
PB understanding	5.67	5.96	4.69	2.04	4.92
Optimism	4.32	5.42	4.28	1.93	3.24
Healthy sleep	5.24	5.68	4.49	2.47	3.97
Economy deterioration	3.12	2.93	4.10	2.17	4.54
Interaction deterioration	1.85	1.85	3.61	1.57	3.68
COVID anxiety	2.91	3.03	4.12	1.61	4.56
COVID sleepless	1.51	1.70	3.49	1.38	3.47
Difficulty living	1.62	1.83	3.68	1.45	3.55
Difficulty work	1.89	2.12	3.82	1.59	3.99
PHQ9 item9	0.05	0.02	0.10	0.24	1.23

* Income Scale: 1 = less than 1M yen, 2 = 2M to 4M yen, 3 = 4M to 6M yen

Table B.2: McNemar and Wilcoxon-signed rank results on cluster transitions with Ambivalent Cluster as source

Feature	Ambivalent	Optimistic	Resilient	Pessimistic	Highly Stressed
Presence of child	changed				
Job group					
Student					
Med self					
Medself COVID					
Med fam					

Medfam COVID					
Current physical		changed	changed		
Past physical					
Current mental					
Past mental					
Current COVID					
Past COVID	changed	changed			
Household income					
Personal income		increased			
Number fam	decreased				
School grade					
Years enrolled					
K6		decreased			increased
PHQ9		decreased			increased
GAD7		decreased			increased
SSS8	increased			decreased	increased
PTGIX	decreased	increased	increased		increased
MAIA 1	increased	increased	increased	decreased	increased
MAIA 2	decreased	decreased	decreased	increased	decreased
MAIA 3				increased	decreased
MAIA 4		increased	decreased	decreased	
MAIA 5		increased	decreased	decreased	increased
MAIA 6		increased	decreased	decreased	
MAIA 7	increased	increased	decreased	decreased	increased
MAIA 8		increased	decreased	decreased	

SHS		increased		decreased	decreased
UCLA		decreased		increased	increased
LSNS6		increased	decreased	decreased	decreased
Frustration			increased	decreased	increased
Exercise		increased	increased	decreased	
Healthy diet		increased		decreased	increased
Favorite activity	increased	increased	increased	decreased	
Interaction offline	increased	increased	increased	decreased	increased
Interaction online		increased	increased	decreased	increased
PB Continuous			decreased	decreased	decreased
PB Altruistic			decreased	decreased	decreased
PB Avoidant			decreased	decreased	
Trust gov		increased		decreased	
Trust social media		increased	increased	decreased	increased
Vaccination will	increased	increased	decreased	decreased	
PB understanding			decreased	decreased	decreased
Optimism		increased		decreased	decreased
Healthy sleep		increased	decreased	decreased	decreased
Economy deterioration			increased	decreased	increased
Interaction deterioration	increased	increased	increased		increased
COVID anxiety	decreased		increased	decreased	increased
COVID sleepless			increased		increased
Difficulty living	decreased		increased	decreased	increased
Difficulty work			increased	decreased	increased
PHQ9 item9		decreased			increased

Table B.3: McNemar and Wilcoxon-signed rank results on cluster transitions with Optimistic Cluster as source

Feature	Ambivalent	Optimistic	Resilient	Pessimistic	Highly Stressed
Presence of child					
Job group					
Student					
Med self					
Medself COVID					
Med fam					
Medfam COVID					
Current physical					
Past physical					
Current mental					
Past mental					
Current COVID					
Past COVID		changed	changed		
Household income					
Personal income					
Number fam	decreased	decreased	decreased		
SchoolGrade					
YearsEnrolled					
K6	increased		increased		increased
PHQ9	increased		increased		increased
GAD7	increased		increased		increased
SSS8		increased	increased	decreased	increased
PTGIX	decreased		decreased	decreased	decreased

MAIA 1	decreased	increased			increased
MAIA 2		decreased			decreased
MAIA 3	decreased		decreased		decreased
MAIA 4	decreased	decreased	decreased	decreased	decreased
MAIA 5	decreased		decreased	decreased	decreased
MAIA 6	decreased	increased	decreased	decreased	decreased
MAIA 7	decreased		decreased	decreased	decreased
MAIA 8	decreased		decreased	decreased	decreased
SHS	decreased		decreased	decreased	decreased
UCLA	increased		increased	increased	increased
LSNS6	decreased		decreased		decreased
Frustration	decreased		increased	decreased	increased
Exercise	decreased		decreased	decreased	decreased
Healthy diet	decreased		decreased	decreased	decreased
Favorite activity	decreased	increased	decreased	decreased	decreased
Interaction offline	decreased	increased		decreased	
Interaction online	decreased		increased	decreased	decreased
PB Continuous			decreased	decreased	decreased
PB Altruistic			decreased	decreased	decreased
PB Avoidant			decreased	decreased	decreased
Trust gov	decreased		decreased	decreased	
Trust social media	decreased			decreased	
Vaccination will		increased	decreased	decreased	decreased
PB understanding		increased	decreased	decreased	decreased
Optimism	decreased	increased	decreased	decreased	decreased

Healthy sleep			decreased	decreased	decreased
Economy deterioration	decreased		increased	decreased	increased
Interaction deterioration			increased		increased
COVID anxiety	decreased	decreased	increased	decreased	increased
COVID sleepless	decreased		increased		increased
Difficulty living	decreased	decreased	increased		increased
Difficulty work	decreased	decreased	increased		increased
PHQ9 item9			increased		increased

Table B.4: McNemar and Wilcoxon-signed rank results on cluster transitions with Resilient Cluster as source

Feature	Ambivalent	Optimistic	Resilient	Pessimistic	Highly Stressed
Presence of child			changed		
Job group					
Student					
Med self					
Medself COVID					
Med fam					
Medfam COVID					
Current physical		changed			
Past physical		changed			
Current mental					
Past mental					
Current COVID					
Past COVID			changed		

Household income		increased			
Personal income		increased			
Number fam	decreased	decreased	decreased	decreased	decreased
School grade					
Years enrolled					
K6		decreased	increased		increased
PHQ9		decreased			increased
GAD7		decreased			increased
SSS8				decreased	increased
PTGIX	decreased			decreased	decreased
MAIA 1		increased	increased	decreased	increased
MAIA 2		decreased	decreased	increased	decreased
MAIA 3	decreased	increased	decreased	increased	decreased
MAIA 4	increased	increased	increased	decreased	increased
MAIA 5	increased	increased	increased	decreased	increased
MAIA 6	increased	increased	increased	decreased	increased
MAIA 7	increased	increased	increased	decreased	increased
MAIA 8	increased	increased	increased	decreased	increased
SHS	increased	increased			decreased
UCLA	decreased	decreased			increased
LSNS6		increased		decreased	decreased
Frustration	decreased	decreased		decreased	increased
Exercise	decreased	increased		decreased	
Healthy diet	decreased	increased		decreased	
Favorite activity	decreased	increased		decreased	decreased

Interaction offline	decreased	increased	increased	decreased	decreased
Interaction online	decreased	decreased		decreased	decreased
PB Continuous	increased	increased		decreased	
PB Altruistic	increased	increased		decreased	
PB Avoidant	increased	increased	decreased	decreased	
Trust gov		increased		decreased	
Trust social media	decreased			decreased	
Vaccination will	increased	increased	increased	decreased	
PB understanding	increased	increased		decreased	
Optimism	increased	increased		decreased	decreased
Healthy sleep	increased	increased		decreased	decreased
Economy deterioration	decreased	decreased		decreased	increased
Interaction deterioration	decreased	decreased		decreased	
COVID anxiety	decreased	decreased	decreased	decreased	increased
COVID sleepless	decreased	decreased		decreased	
Difficulty living	decreased	decreased	decreased	decreased	
Difficulty work	decreased	decreased		decreased	
PHQ9 item9					increased

Table B.5: McNemar and Wilcoxon-signed rank results on cluster transitions with Pessimistic Cluster as source

Feature	Ambivalent	Optimistic	Resilient	Pessimistic	Highly Stressed
Presence of child					
Job group					
Student					

Med self					
Medself COVID					
Med fam					
Medfam COVID					
Current physical					
Past physical					changed
Current mental					
Past mental					
Current COVID					
Past COVID					
Household income					
Personal income					
Number fam	decreased		decreased		decreased
SchoolGrade			increased		
YearsEnrolled			increased		
K6	increased				increased
PHQ9					increased
GAD7					increased
SSS8	increased		increased		increased
PTGIX	increased	increased	increased	increased	increased
MAIA 1	increased	increased	increased		increased
MAIA 2	decreased	decreased	decreased		decreased
MAIA 3	decreased		decreased		decreased
MAIA 4	increased	increased	increased	increased	increased
MAIA 5	increased	increased	increased	increased	increased

MAIA 6	increased	increased	increased		increased
MAIA 7	increased	increased	increased		increased
MAIA 8	increased	increased	increased		increased
SHS	increased	increased		decreased	
UCLA		decreased			
LSNS6	increased		increased		
Frustration	increased	increased	increased		increased
Exercise	increased	increased	increased		increased
Healthy diet	increased	increased	increased		increased
Favorite activity	increased	increased	increased		increased
Interaction offline	increased	increased	increased	increased	increased
Interaction online	increased	increased	increased		increased
PB Continuous	increased	increased	increased	increased	increased
PB Altruistic	increased	increased	increased	increased	increased
PB Avoidant	increased	increased	increased		increased
Trust gov	increased	increased	increased		increased
Trust social media	increased	increased	increased		increased
Vaccination will	increased	increased	increased		increased
PB understanding	increased	increased	increased		increased
Optimism	increased	increased	increased		increased
Healthy sleep	increased	increased	increased	increased	increased
Economy deterioration	increased	increased	increased		increased
Interaction deterioration	increased	increased	increased		increased
COVID anxiety	increased	increased	increased		increased
COVID sleepless	increased	increased	increased	increased	increased

Difficulty living	increased	increased	increased	increased	increased
Difficulty work	increased	increased	increased	increased	increased
PHQ9 item9	decreased	decreased			

Table B.6: McNemar and Wilcoxon-signed rank results on cluster transitions with Highly Stressed Cluster as source

Feature	Ambivalent	Optimistic	Resilient	Pessimistic	Highly Stressed
Presence of child					
Job group			changed		
Student					
Med self					
Medself COVID					
Med fam					
Medfam COVID					
Current physical					
Past physical					
Current mental			changed	changed	
Past mental				changed	
Current COVID					
Past COVID					changed
Household income					
Personal income				increased	
Number fam	decreased		decreased		decreased
School grade					
Years enrolled					
K6	decreased	decreased	decreased	decreased	increased
PHQ9	decreased	decreased	decreased	decreased	
GAD7	decreased	decreased	decreased	decreased	increased

SSS8	decreased	decreased	decreased	decreased	increased
PTGIX	decreased	increased	increased	decreased	
MAIA 1	decreased	decreased	decreased	decreased	increased
MAIA 2	increased	increased	increased	increased	decreased
MAIA 3	increased	increased	increased	increased	
MAIA 4		increased	decreased	decreased	
MAIA 5		increased	decreased	decreased	
MAIA 6		increased	decreased	decreased	increased
MAIA 7		increased	decreased	decreased	
MAIA 8		increased	decreased	decreased	increased
SHS	increased	increased	increased	increased	increased
UCLA	decreased	decreased	decreased	decreased	decreased
LSNS6	increased	increased	increased	decreased	decreased
Frustration	decreased	decreased	decreased	decreased	
Exercise		increased	increased	decreased	
Healthy diet		increased	increased	decreased	
Favorite activity		increased	increased	decreased	
Interaction offline		increased	increased	decreased	increased
Interaction online	decreased	increased	increased	decreased	
PB Continuous	increased	increased		decreased	
PB Altruistic	increased	increased		decreased	
PB Avoidant			decreased	decreased	
Trust gov		increased		decreased	
Trust social media		increased	increased	decreased	
Vaccination will	increased	increased	increased	decreased	increased
PB understanding	increased	increased		decreased	
Optimism	increased	increased	increased	decreased	
Healthy sleep	increased	increased	increased	decreased	
Economy deterioration	decreased	decreased		decreased	

Interaction deterioration	decreased	decreased		decreased	increased
COVID anxiety	decreased	decreased	decreased	decreased	decreased
COVID sleepless	decreased	decreased		decreased	
Difficulty living	decreased	decreased		decreased	decreased
Difficulty work	decreased	decreased		decreased	
PHQ9 item9		decreased	decreased	decreased	

Appendix C

Environment and Source Code

C.1 Environment

Listed below are the versions of Python packages used in the project. Note that Google Colab with Python 3.10.12 was used to execute these packages.

- pandas==1.5.3
- numpy==1.23.5
- scipy==1.11.4
- scikit-learn==1.2.2
- hmmlearn==0.3.0
- plotly==5.15.0
- statsmodels==0.14.1
- netgraph==4.13.2

C.2 Sections of the Python Notebook

Loading data

Loads data from a Google Spreadsheet. Separates the data by survey then renames their columns to a common name for prediction and comparison purposes.

Data preprocessing

Drops extra columns in Surveys 2 to 4 then imputes missing columns with average values from Survey 5. One-hot encodes categorical variables and standardize numerical variables. Outliers are dropped based on household and personal income.

Training and prediction of coping profiles using k-means clustering

K-means clustering model was fitted using Survey 5 then the fitted model was used to predict clusters in Surveys 2 to 4. Radio plots were graphed to visualize similarities and differences in the defining features.

Estimating coping profile transition probability matrix by training Hidden Markov Model on Surveys 2 to 5

Hidden Markov Model (HMM) was cross-validated using stratified shuffle split with five iterations. Samples were generated by the HMM and predicted using the fitted k-means clustering model. Radio plots were graphed to compare with the results from the k-means clustering model.

Statistical Tests

Kruskal-Wallis test was applied on the same clusters but different times. McNemar

and one-tailed Wilcoxon signed-rank tests were performed on samples who experienced coping file transitions.

Computing values for the probabilistic graphs

Simulation of initial state transition probabilities to produce stationary distributions.

Plotting probabilistic graphs

Plotting of state transition matrices into probabilistic graphs.

C.3 Helper Functions

`preprocess_survey`

Function to run data preprocessing per survey.

`radio_plot`

Function to generate radio plots of the defining features from the results of the k-means clustering and HMM.

`map_hidden_state_to_coping_profile`

Function to match a hidden state to its corresponding coping profile using the values of its defining features. Note that the hidden states learned by the HMM does not have a uniform order.

`stratified_shuffle_split`

Function to perform stratified shuffle split using five iterations and output classification metrics.

plot_features

Function to generate box plots of the defining features of the same clusters but different times.

distribution_statistical_tests

Function to perform Kruskal-Wallis test on the defining features for a single survey.

find_state_transitions

Function to find all state transitions for a certain coping profile.

paired_statistical_tests

Function to perform McNemar and one-tailed Wilcoxon signed-rank tests.

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